

# Scanning Electron Microscopy



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4.1 Scanning Electron Microscopy Principles

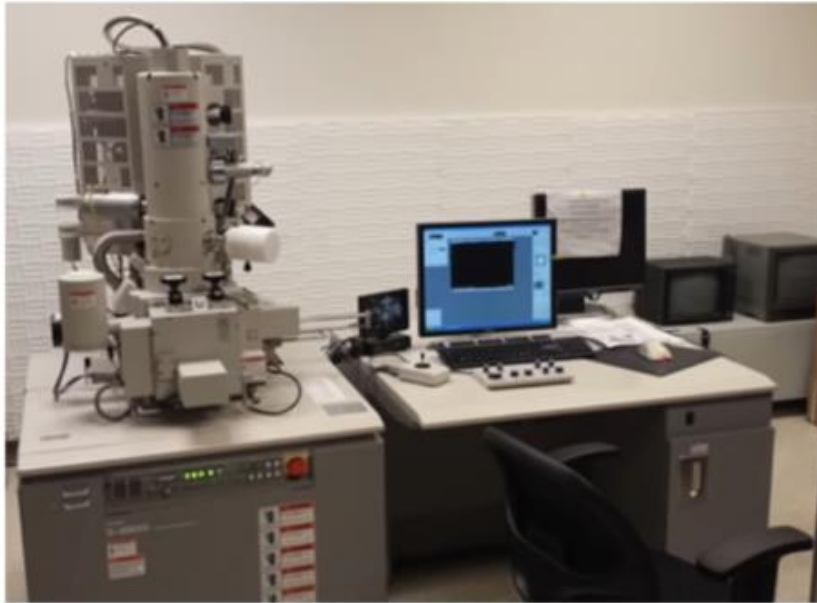
4.2 Electron Beam–Specimen Interaction

4.3 SEM Operating Parameters

4.4 Applications



# What is an Scanning Electron Microscope?

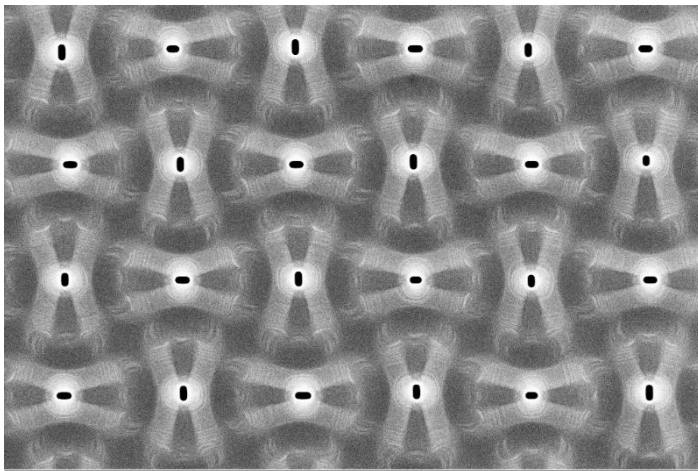


MRL Hitachi S4800

- A **Scanning Electron Microscope (SEM)** is an instrument for **observing** and **analyzing** the surface microstructure of a bulk sample using a **finely focused beam of electrons**
- SEMs can achieve 1-4 nm imaging resolution depending on the instrument

## Primary Applications

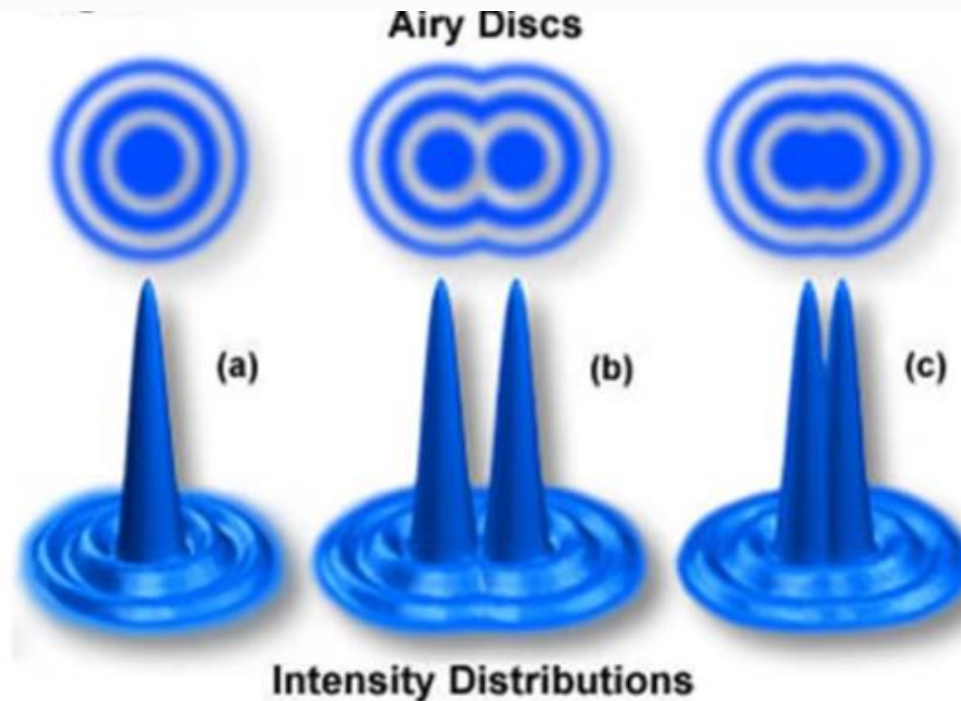
- **Surface topography / morphology**
- **Composition analysis**
- **Crystallography**
- **Optical / Electronic properties**
- **and more ...**



SEM image of Ge on Si from my lab



# Resolution-what is it?



- Ability to tell the difference between two close objects vs. one big object
- Electron beam is not a point → Gaussian with nanoscale width
  - Diffraction causes the 'Airy' disc
  - Rayleigh Criterion: Two points are resolvable if the maximum of the first disc coincides with the minimum of the second

# What are the Advantages of Electrons?

## Resolution limits (diffraction)

Abbe equation: 
$$d = \frac{0.61\lambda}{n \sin \alpha}$$

$d$  – resolution

$\lambda$  – wavelength of particle

$n$  – refractive index relative to free space

$\alpha$  – semi-angle in radians

| Microscopy | $\lambda$ (nm) | $n \sin \alpha$ | $d$ (nm)   | Mag.*  |
|------------|----------------|-----------------|------------|--------|
| Light      | 400            | 1               | <b>200</b> | 1000   |
| Electron   | 0.3            | 0.015           | <b>0.3</b> | 500000 |

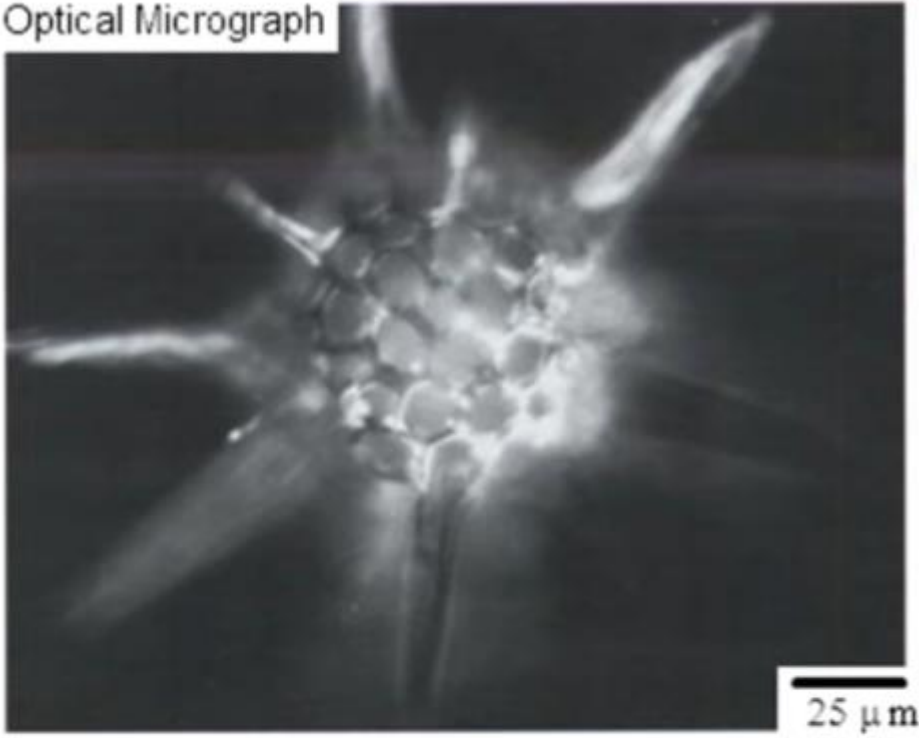
This is the best (approximate) possible resolution. In practice the resolution will be worse...

\*Practical mag. =  $0.2\text{mm (eye)}/d$



# A General Comparison: Optical vs. SEM Imaging

Optical Micrograph



Optical



SEM (secondary electron)

Compared to optical imaging, SEM gives:

- Better resolution
- Better depth of field (how much is in focus)



# 4.1 Scanning Electron Microscopy Principles

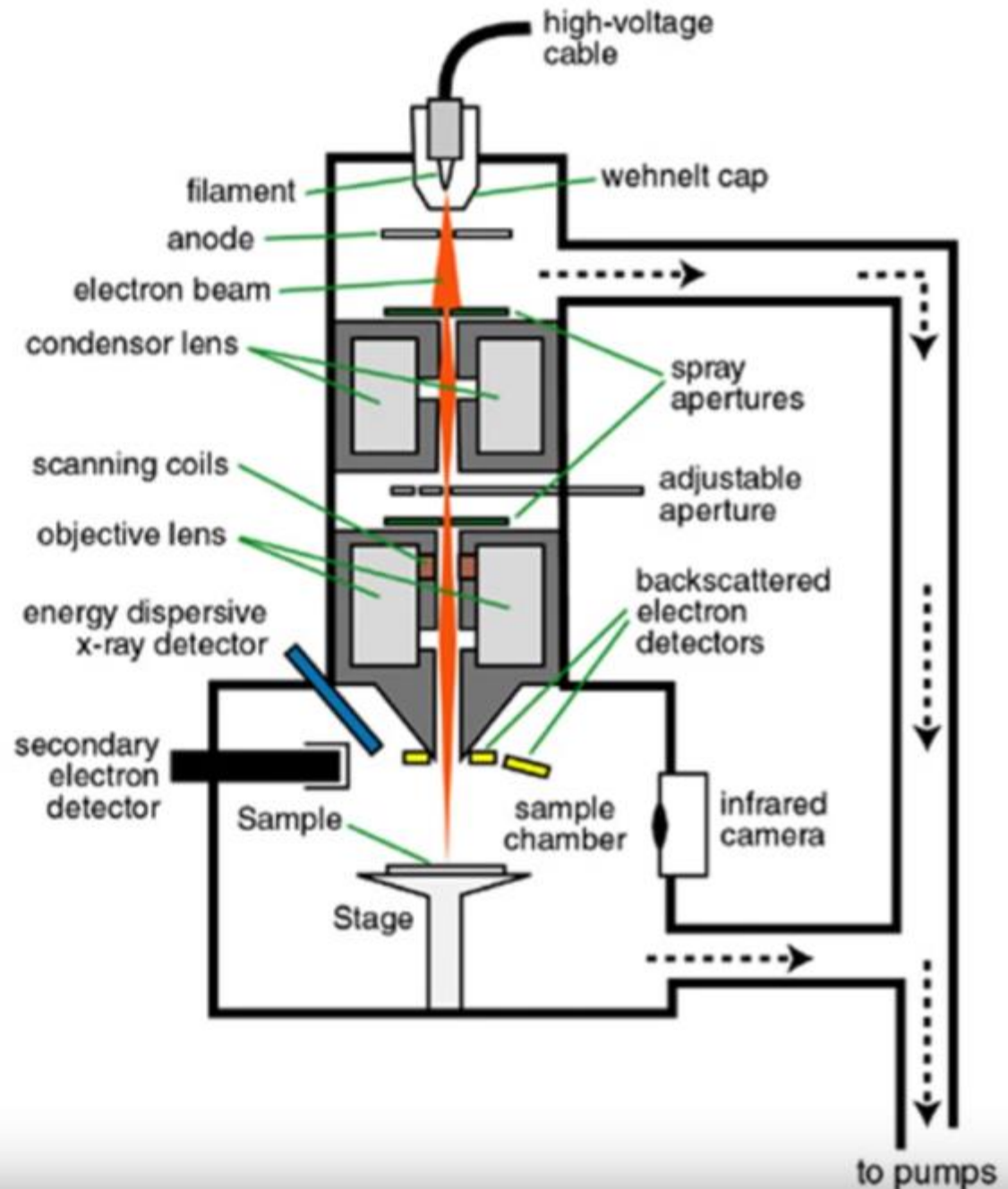
- ◆ 4.1.1 The Electron Gun
- ◆ 4.1.2 The Condenser and Objective Lenses
- ◆ 4.1.3 Scanning Coils
- ◆ 4.1.4 Specimen Chamber





# SEM Component Breakdown

- Vacuum system
- Electron source (gun)
- Optics
  - Lens
  - Aperture
  - Deflectors (导流板)
- Scanning system
- Detectors





# A General Look at Electron Sources

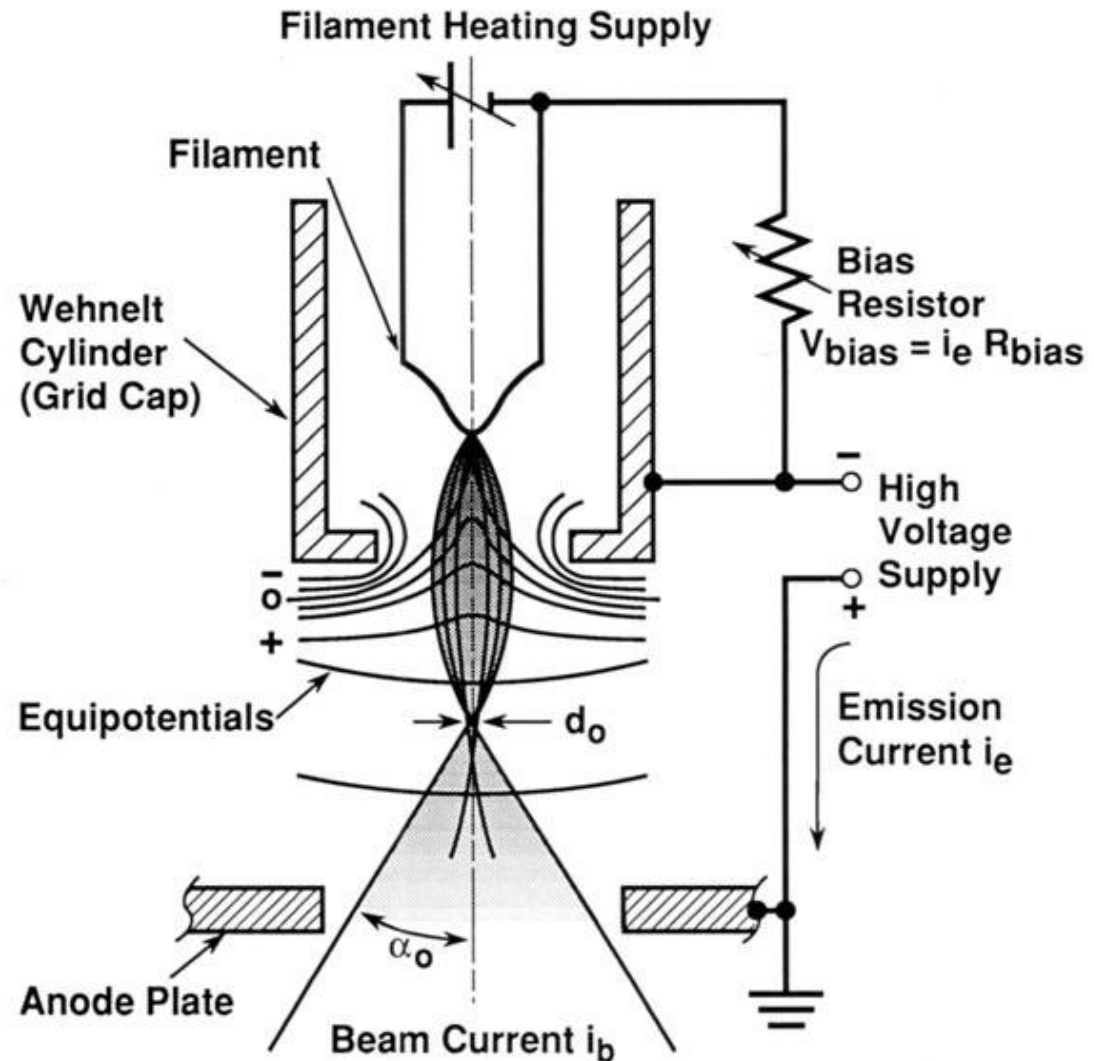
## Purpose:

- Produce electrons
- Roughly shape the beam
- Set the electron energy
- (Accelerating Voltage)

## Two source types:

- Thermionic (heat)
- Field emitter (electric field)

Each source has advantages and disadvantages

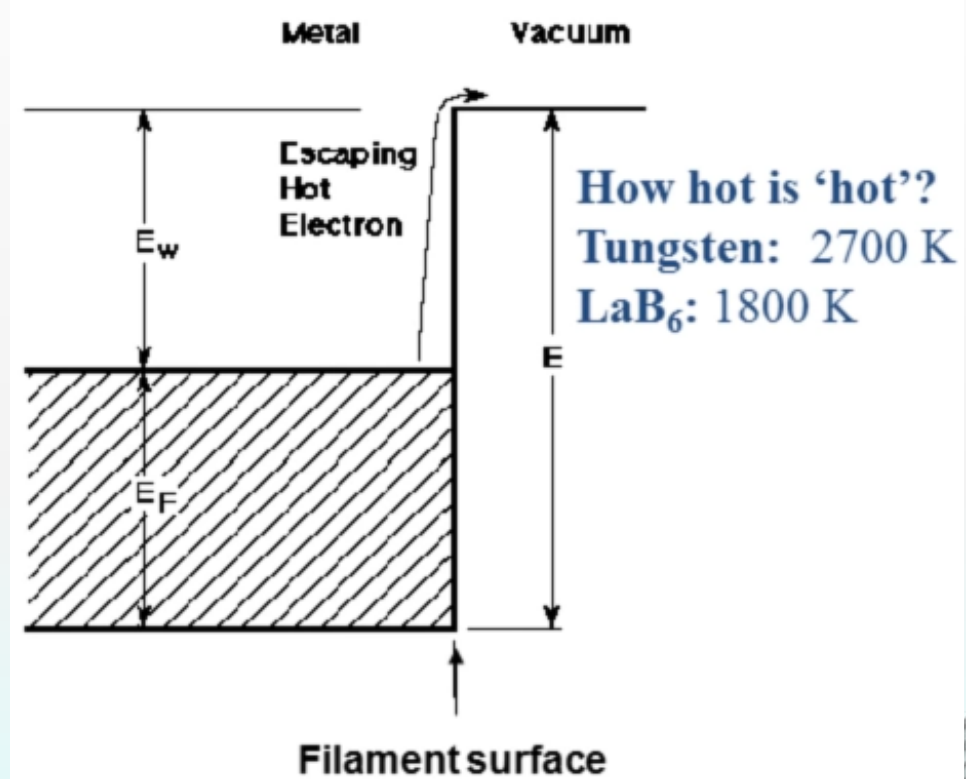
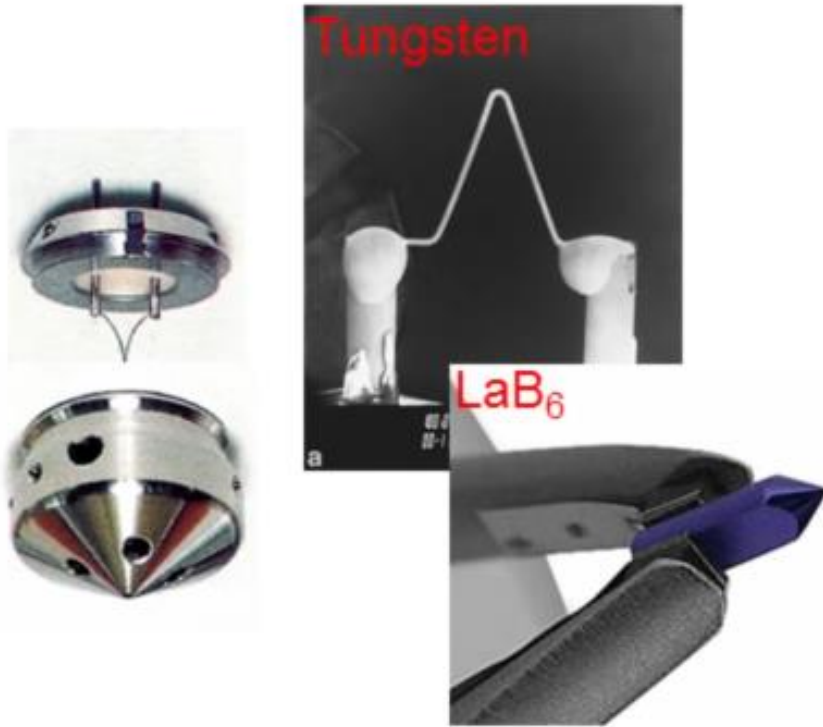


**Figure 2.4.** Schematic diagram of the conventional self-biased thermionic (triode) electron gun (adapted from Hall, 1966).

# Electron Sources-Thermionic (Fancy Lightbulb)

## Tungsten and LaB6 Filaments

Filaments are heated, allowing electrons to overcome the work function

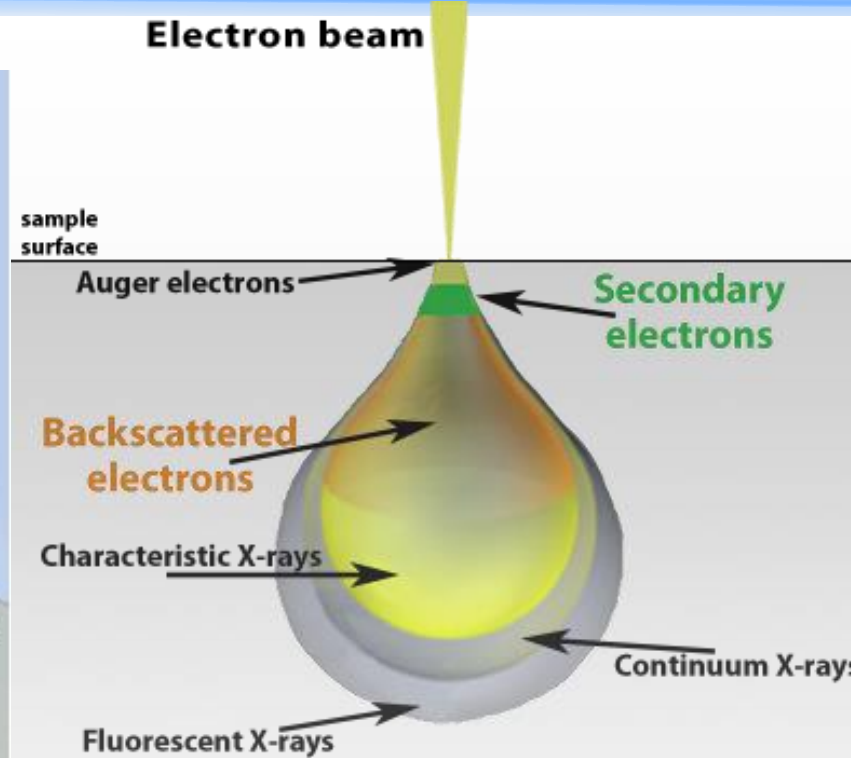
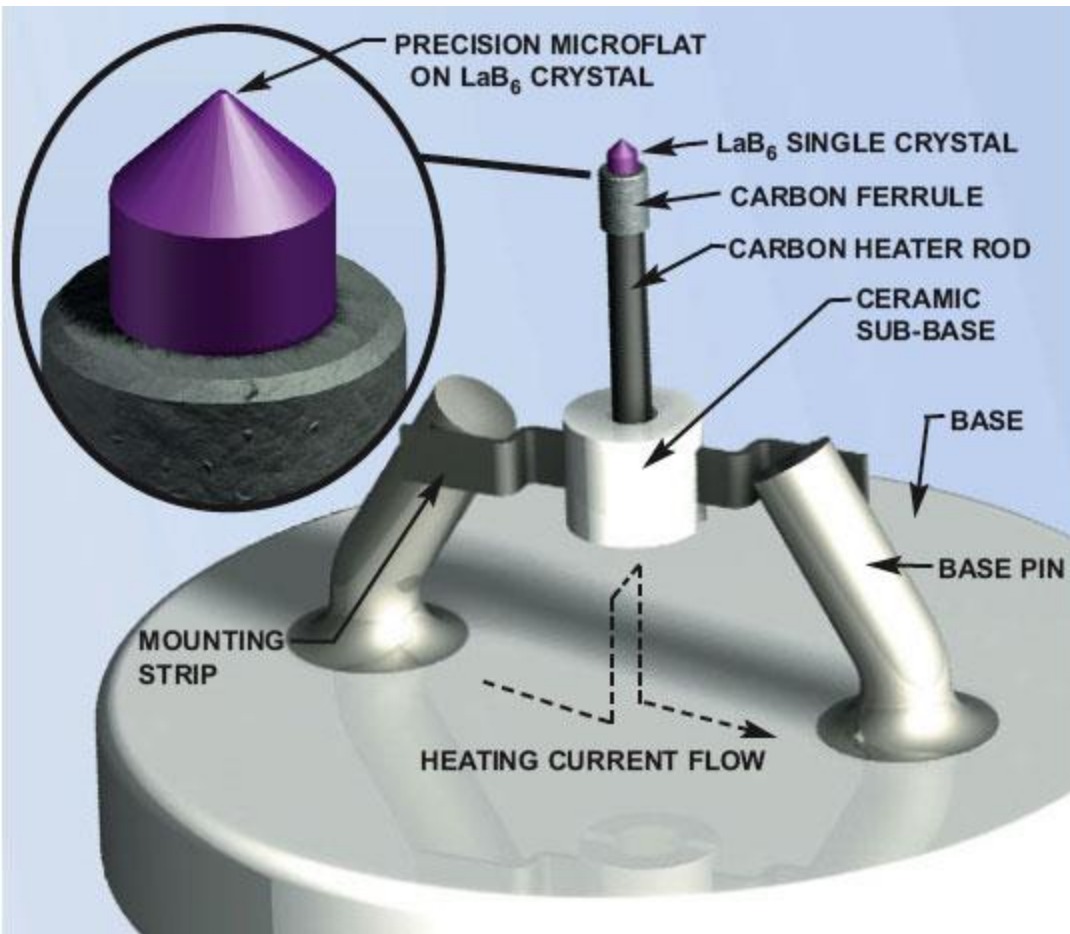


### Major Advantages:

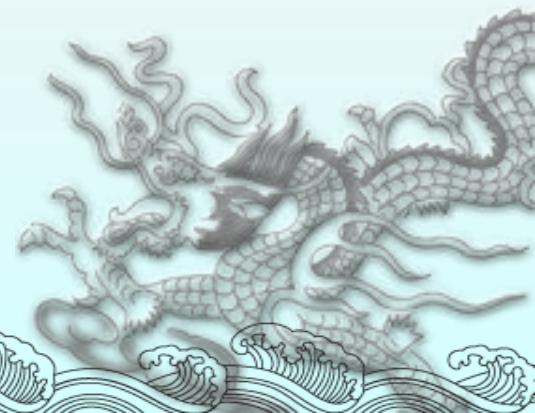
- Very high probe current obtainable
- Stable current
- Less complex vacuum system

### Disadvantages:

- Lower resolution
- Relatively short lifetimes



**Heating current path through precision-machined, single-piece carbon rod and mounting strips; sub-base provides rigidity and easier mounting.**

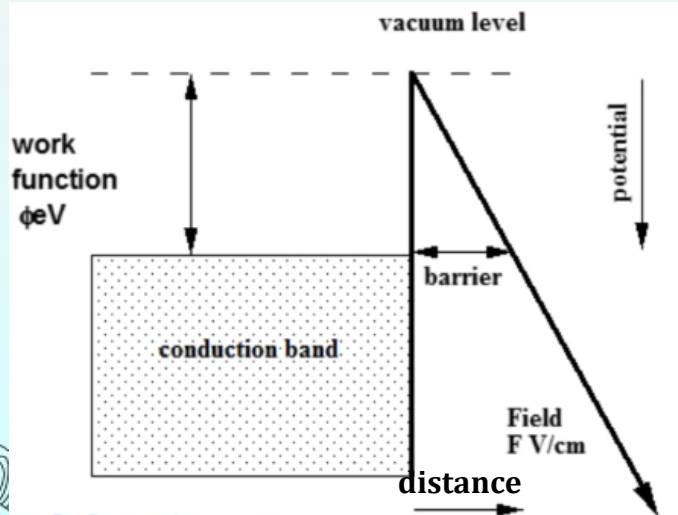
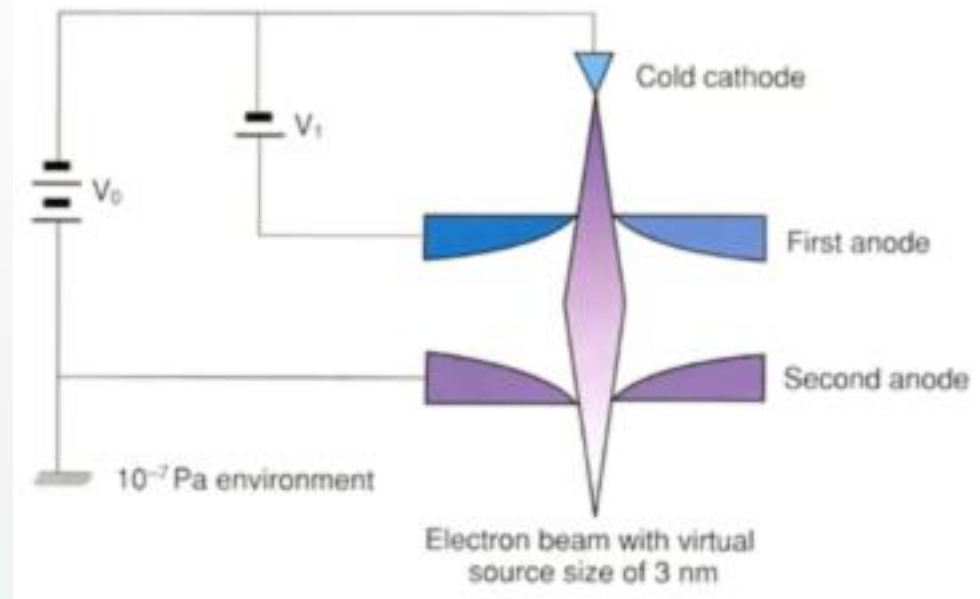
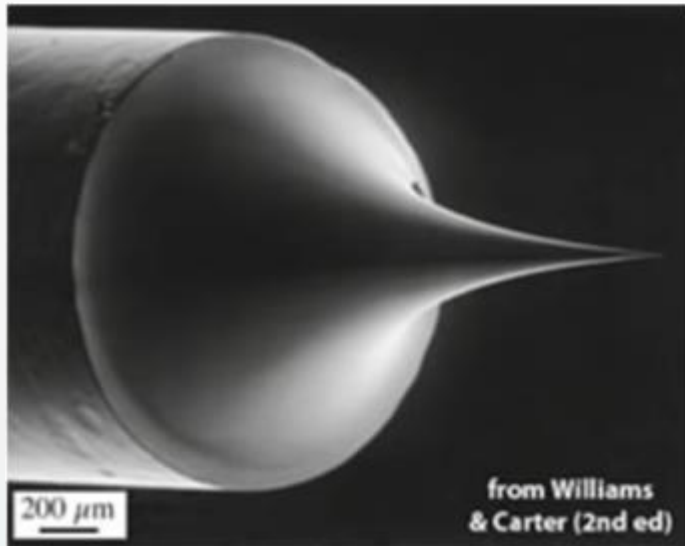




# Electron Sources-Cold Field Emitter (Resolution)

## Sharp Single Crystal (310) Tungsten Tip

An electric field is applied to a sharp tip, electrons are stripped via tunneling (no heating!)



### Major Advantages:

- High resolution (best?)
- Very long potential source lifetime

### Disadvantages:

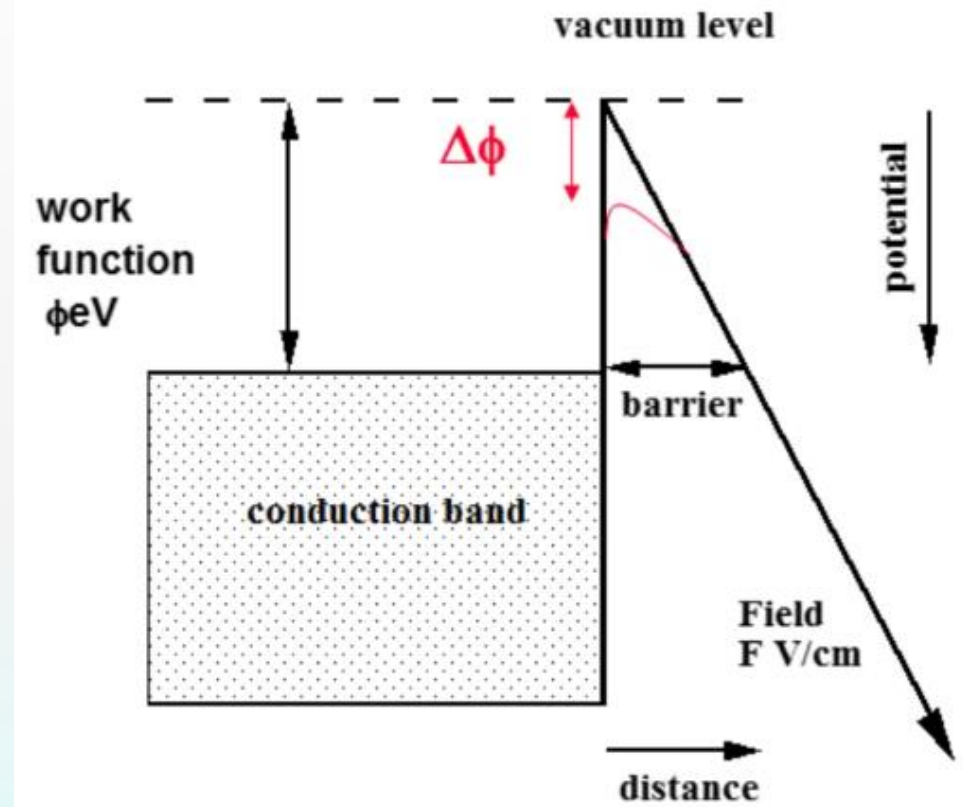
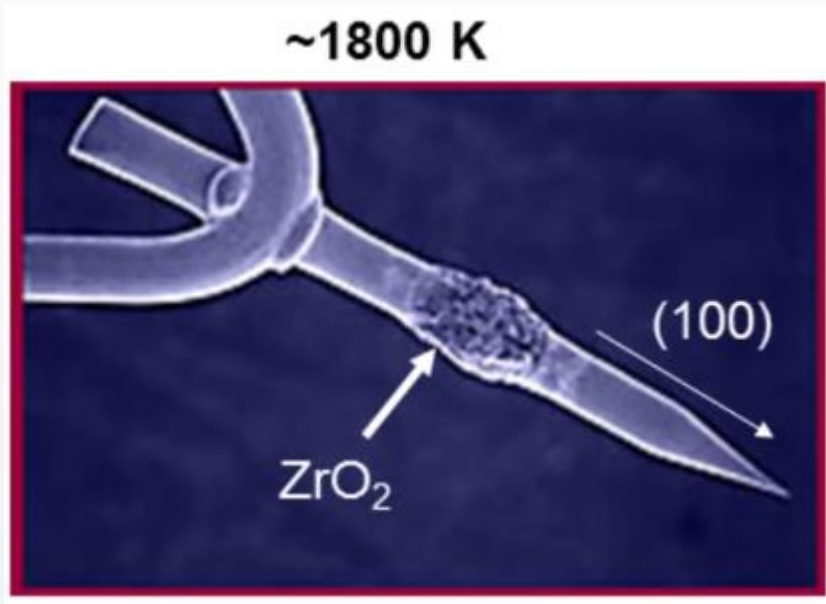
- Lowest maximum probe current
- Poor short and long term probe current stability
- Needs periodic "flashing"
- Requires ultra-high vacuum in gun area



# Electron Sources-Thermal Field (Schottky) Emitter

## Sharp Single Crystal (100) Tungsten Tip with $\text{ZrO}_2$ Film

Combined heating and electric field (hybrid source)



### Major Advantages:

- High resolution
- High probe currents obtainable
- Long potential source lifetime
- Excellent current stability

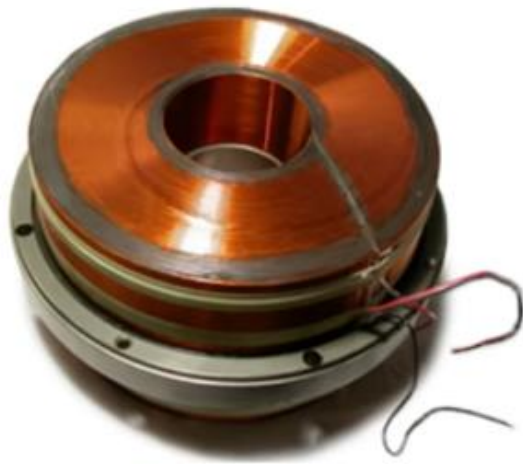
### Disadvantages:

- Requires ultra-high vacuum in gun area
- Finite lifetime

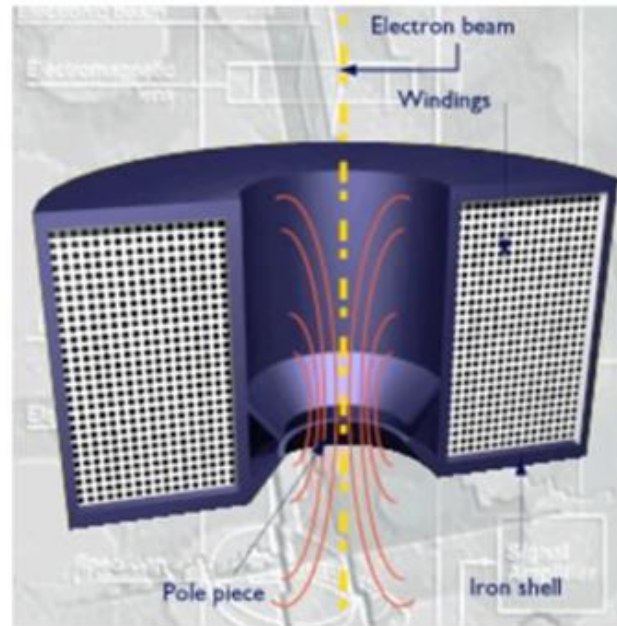
# **Lenses and their Aberrations**



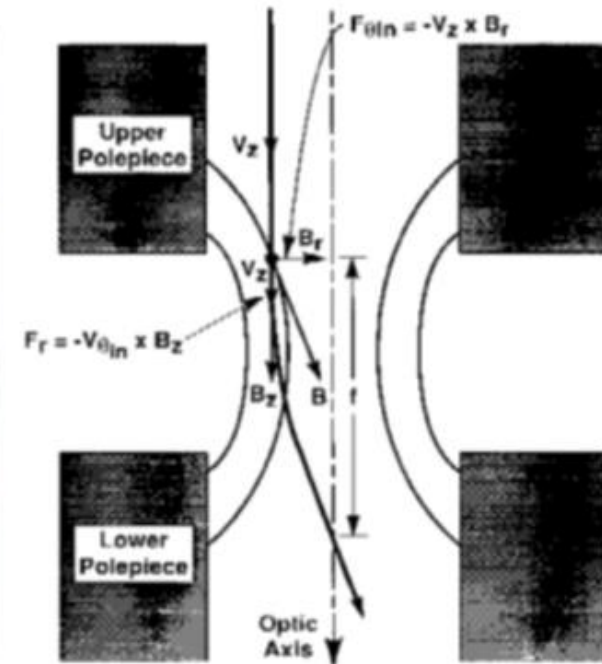
# The Electromagnetic Lens



An electromagnetic lens:  
Wound copper coil  
around soft iron (pole-  
piece), applied current  
produces a magnetic field



Schematic x-section:  
Small slit in iron allows  
mag-field to leak into the  
column, current dictates  
field strength

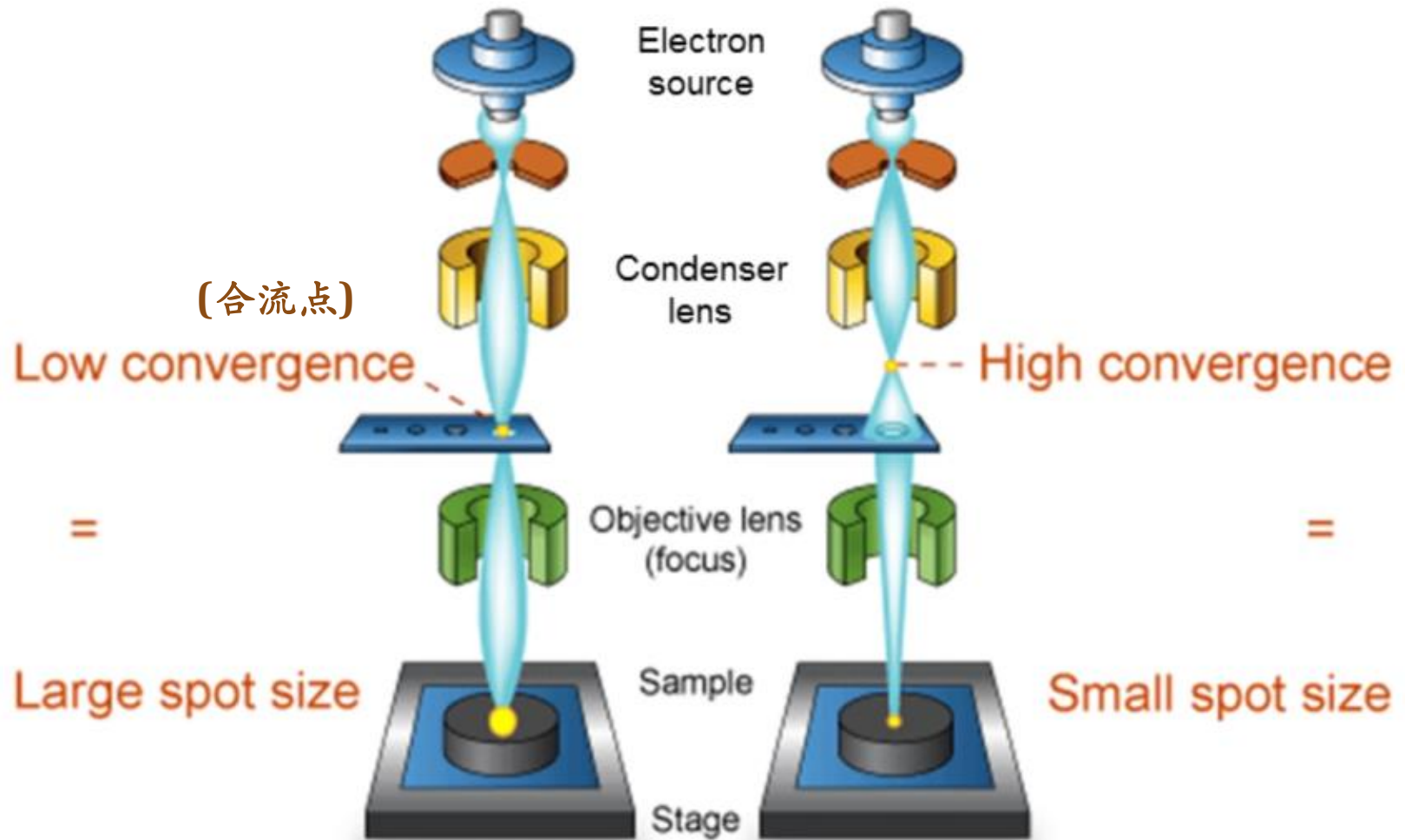


Focusing action:  
Electron trajectories  
changed due to Lorentz  
force  $F = q[E + (v \times B)]$



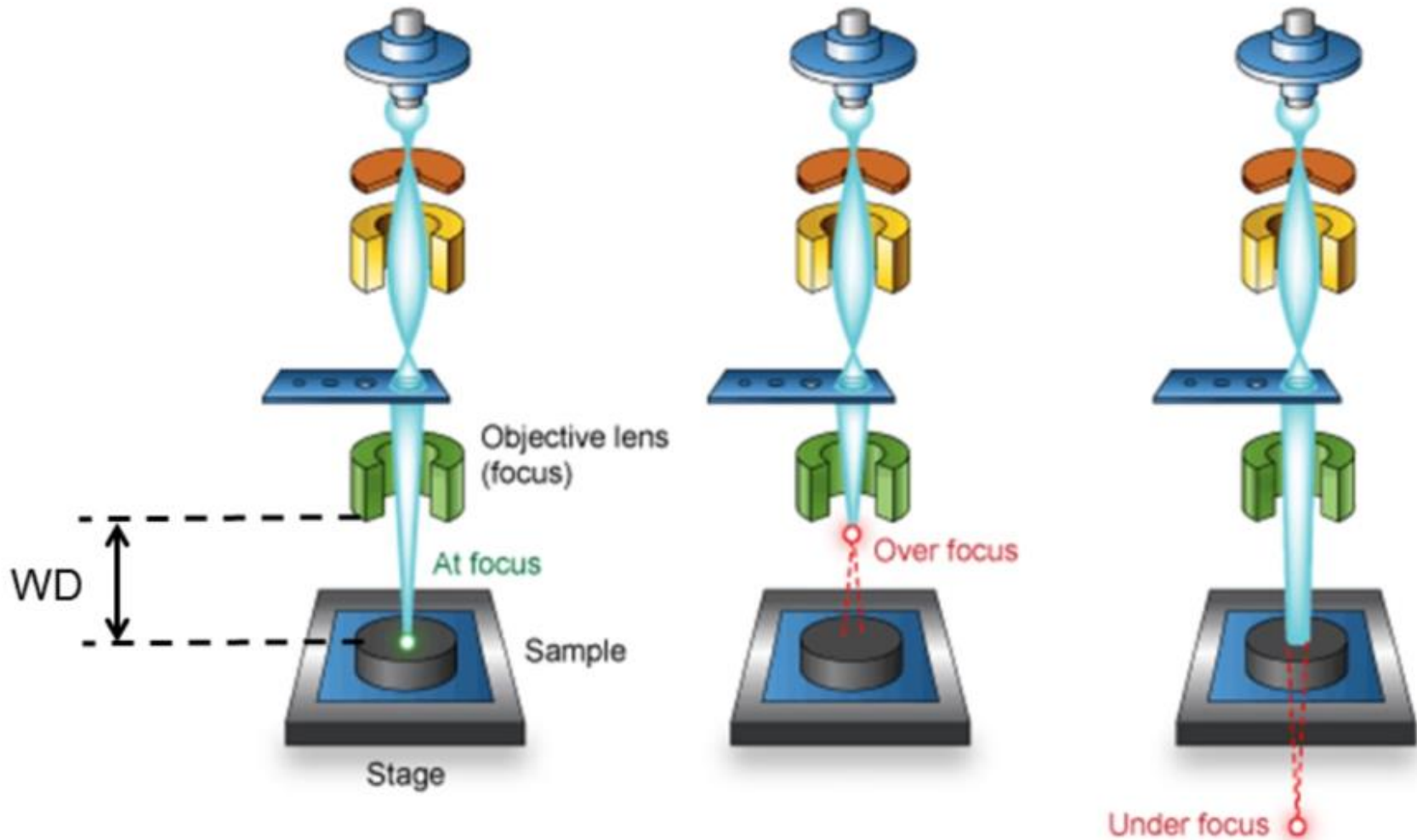


# Function of the Condenser Lens: Spot Size



- Low convergence – more beam accepted by aperture
  - More current, less resolution
- High convergence – less beam accepted by aperture
  - Less current, higher resolution

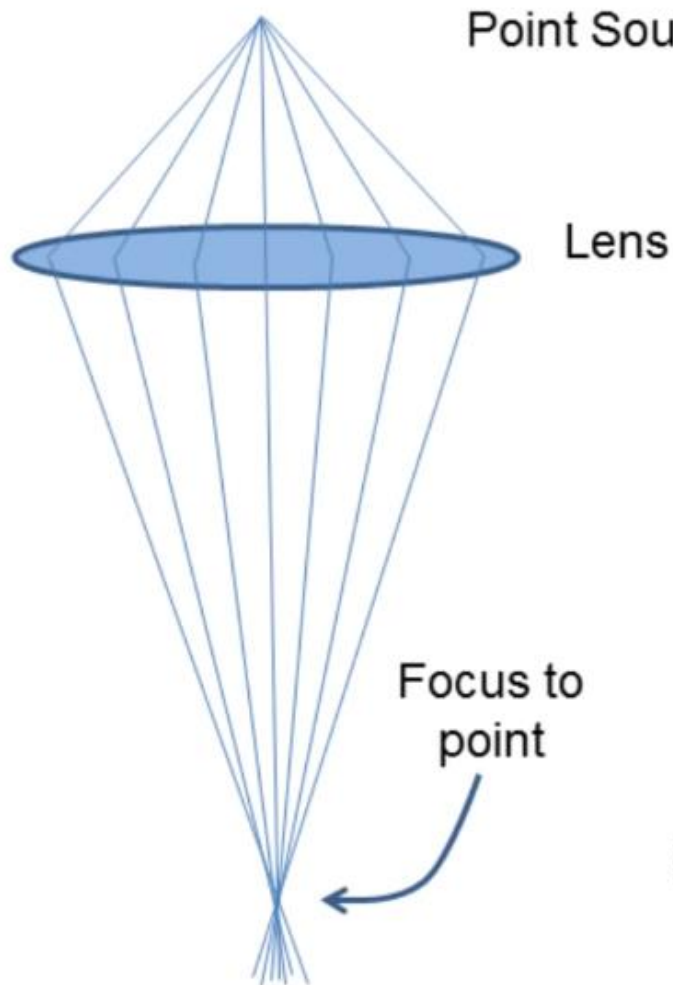
# Function of the Objective Lens: Focus



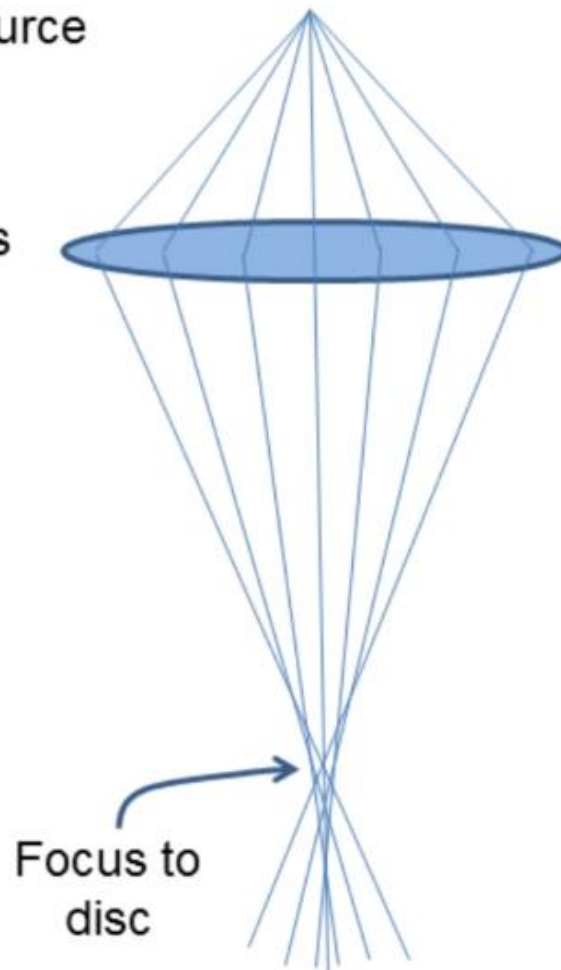
- Sets the focal distance – similar to Z knob on an optical microscope
- Commonly called the Working Distance (WD)

# Rays Ideally are Focused to Same Plane...

Ideal Lens



Actual Lens



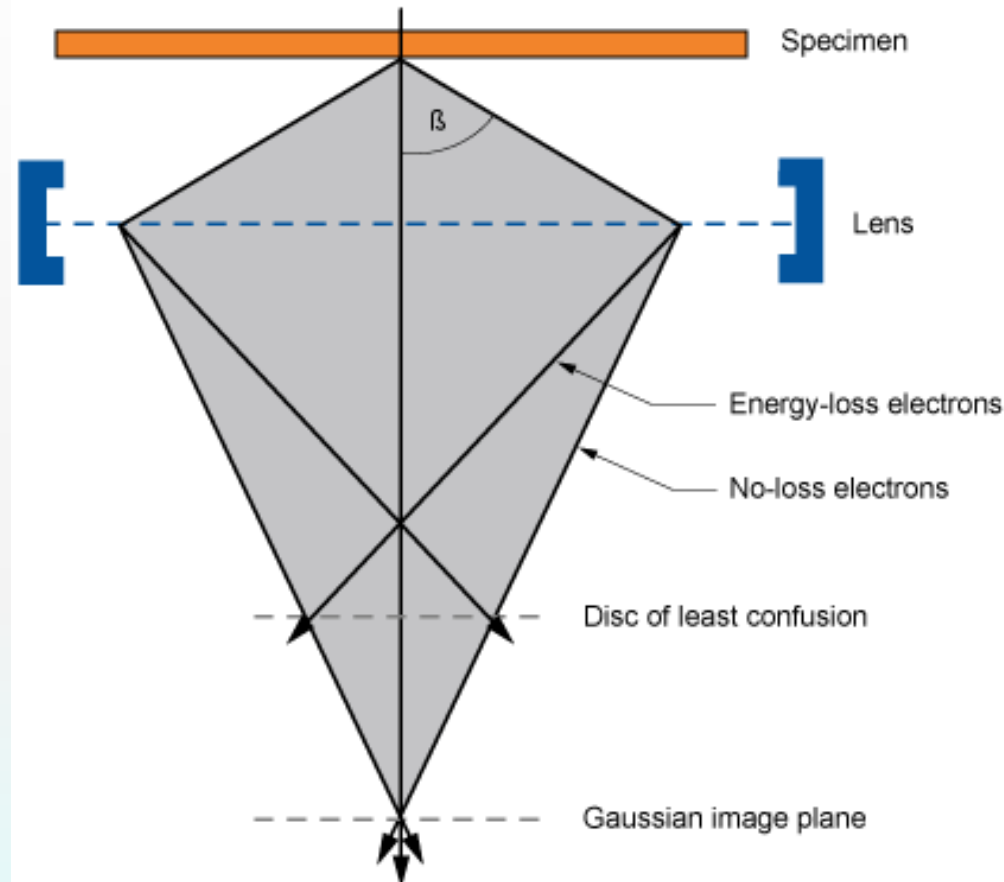
## *Limiting Parameters*

- Spherical Aberration
- Chromatic Aberration
- Aperture diffraction
  - **Astigmatism**

These aberrations reduce resolution!



# Aberration from Electron Source



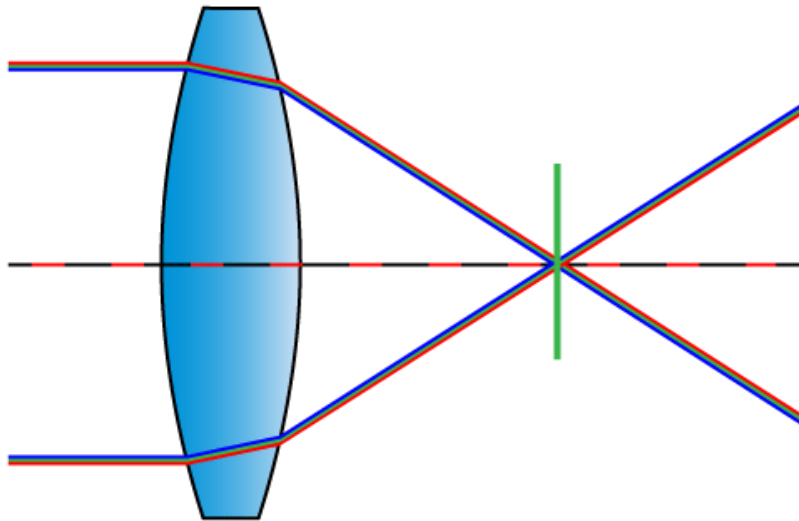
## Chromatic Aberration

色差

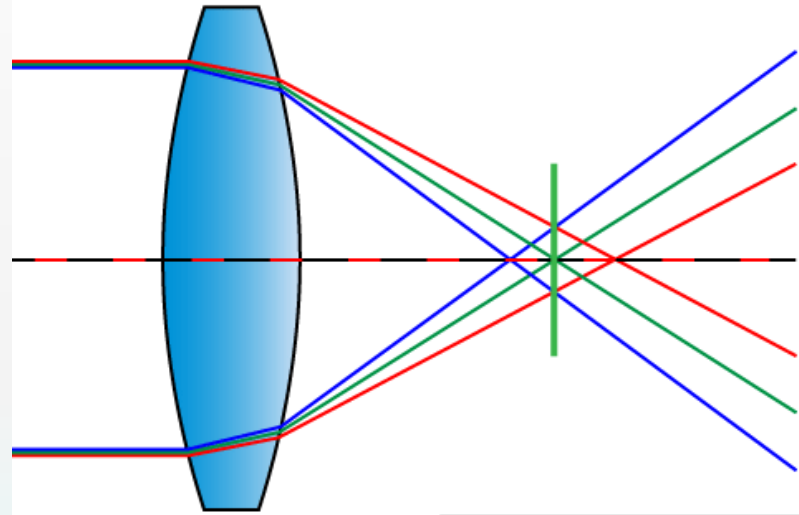
- Electrons with different energy (velocity) are focused to different points in the image plane
- Increases with source energy spread (heated sources!)
- Decreases with increasing electron energy ( $E_0$ )

# Chromatic Aberration

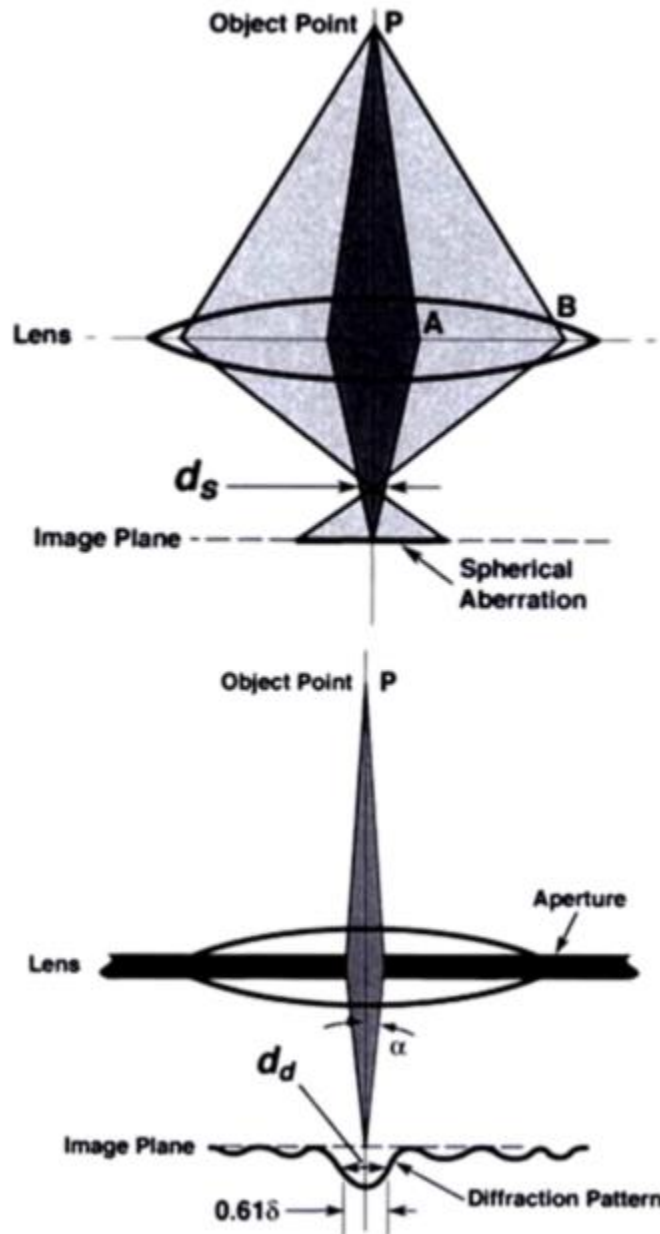
Perfect Lens with no Chromatic Aberration



Longitudinal / Axial Chromatic Aberration



# Aberration from Lenses and Apertures



## Spherical Aberration (球差)

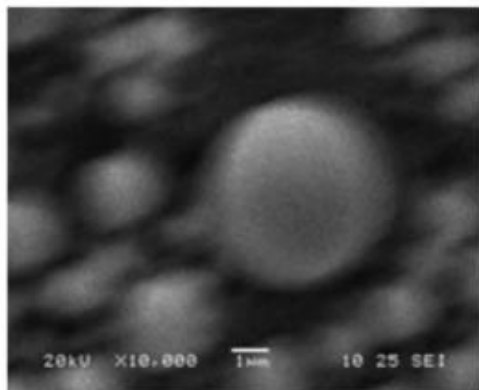
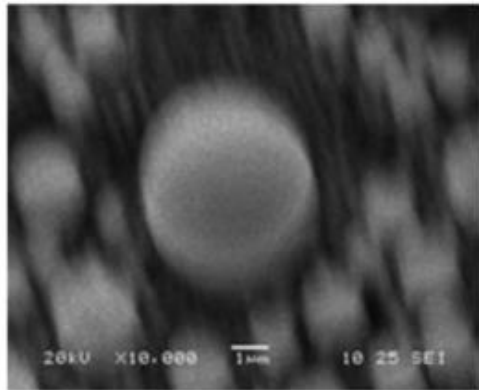
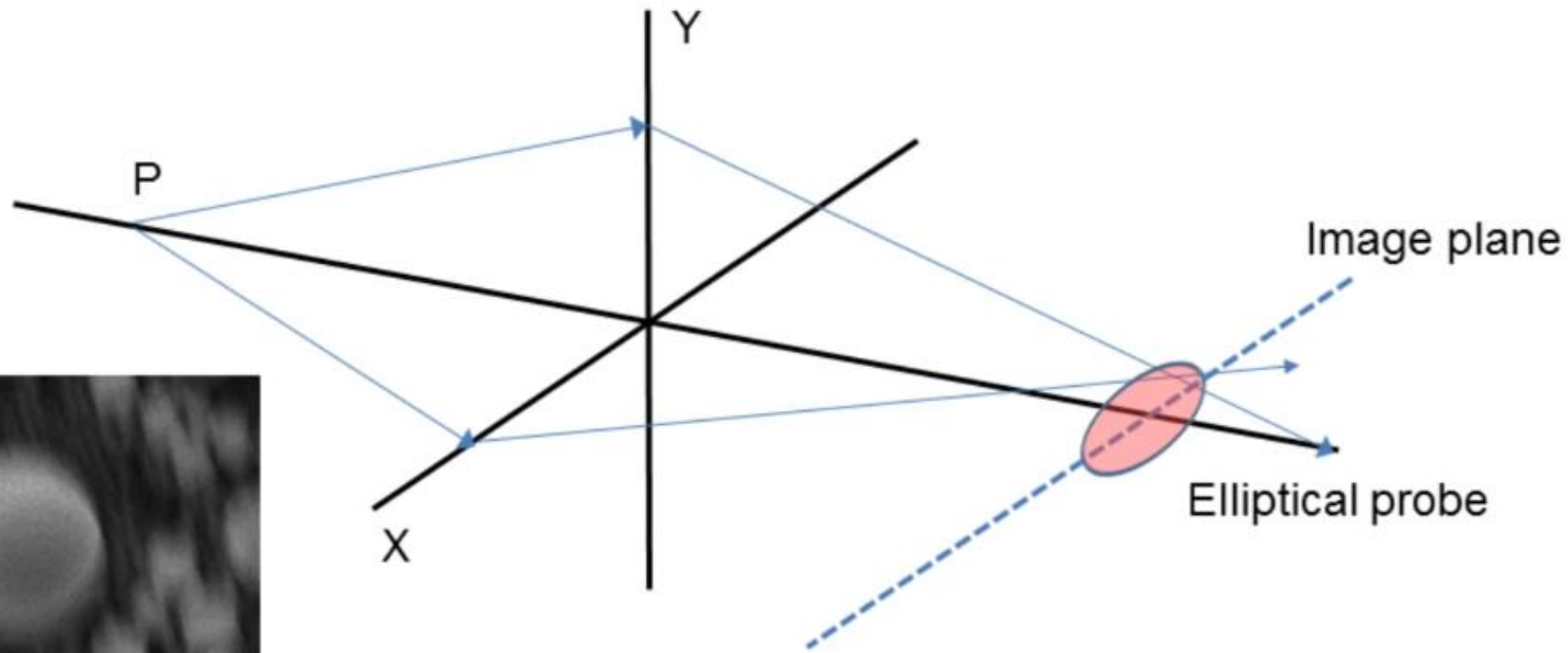
- Electromagnetic lens (mag-field) are not uniform
  - Stronger towards outside
  - Weaker towards center
- Off-axis rays are overfocused
- Solution! Add a limiting aperture

## Aperture Diffraction

- Diffraction effects increase with smaller apertures
- Optimum aperture sizes determined by manufacturer



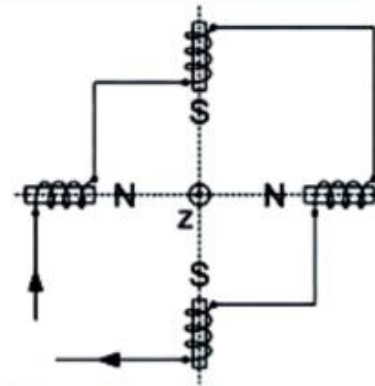
# Last Major Aberration-Astigmatism (像散)



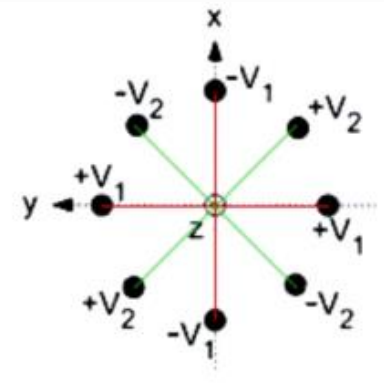
- Caused by slight imperfections in the manufacturing of pole-pieces and the copper windings
- Stray magnetic fields (杂散磁场)
- Images will appear 'stretched' as the operator changes focus

# Shaping the Beam-Deflectors

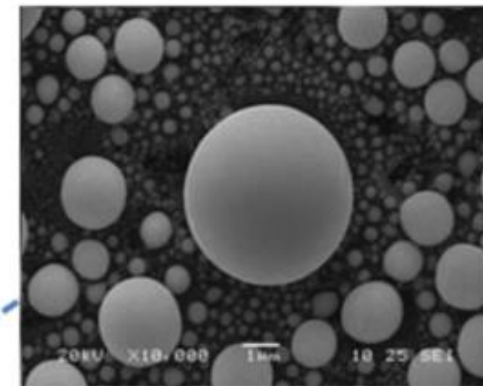
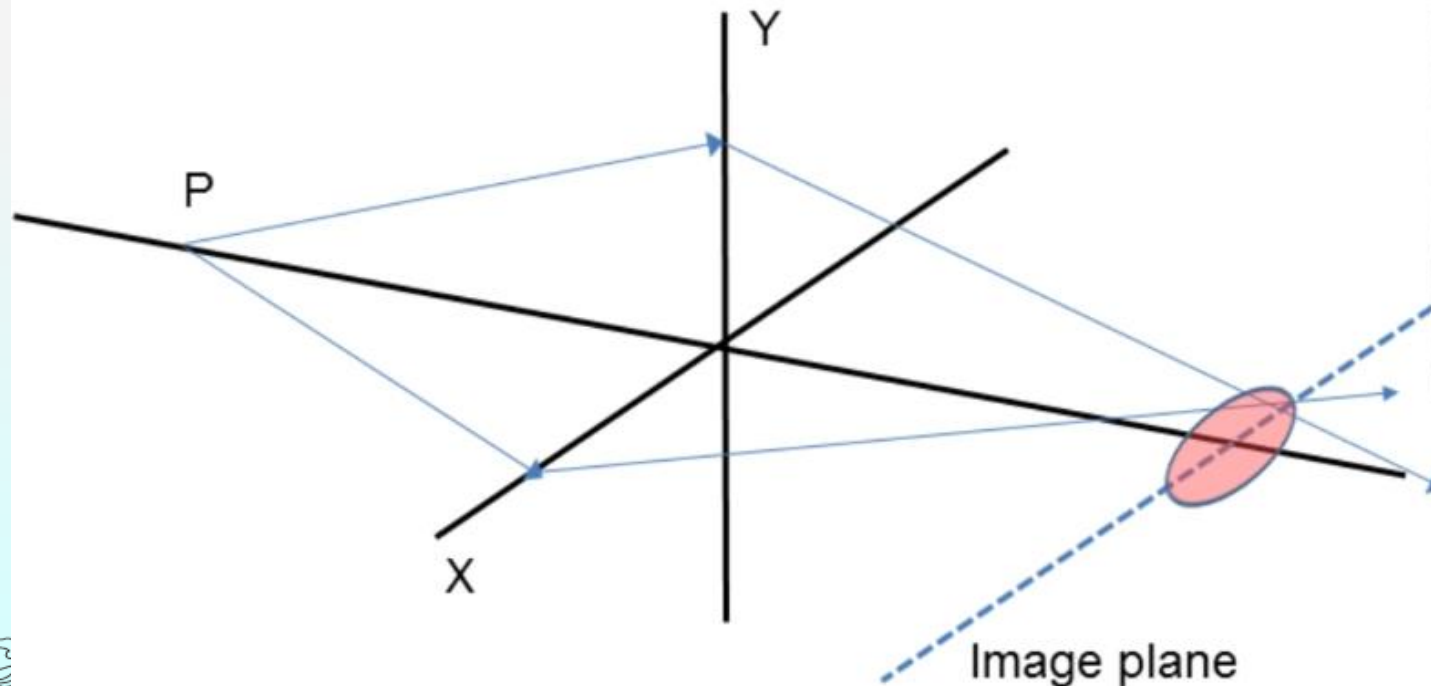
- Electromagnetic lenses perpendicular to the optic axis  
→ Push and pull the beam
- Multiple deflectors are needed to correct for astigmatism



**Quadrupole** lens is basis of a stigmator



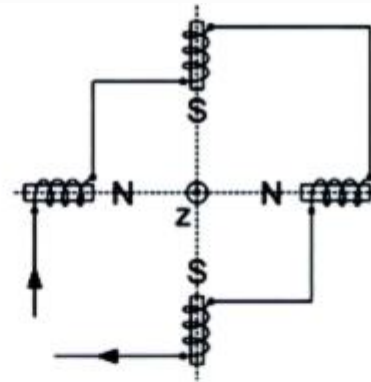
Modern SEMs use **octupole** lens stigmator



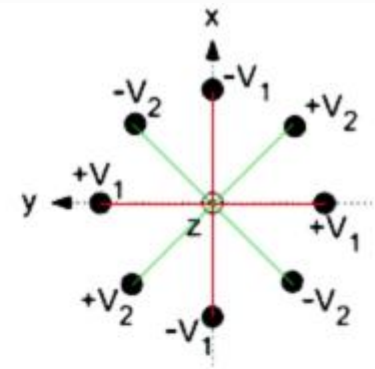
Elliptical probe

# Shaping the Beam-Deflectors

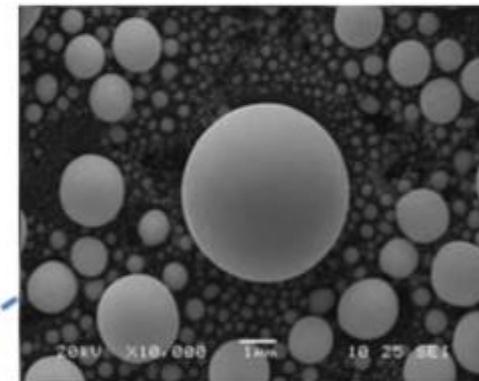
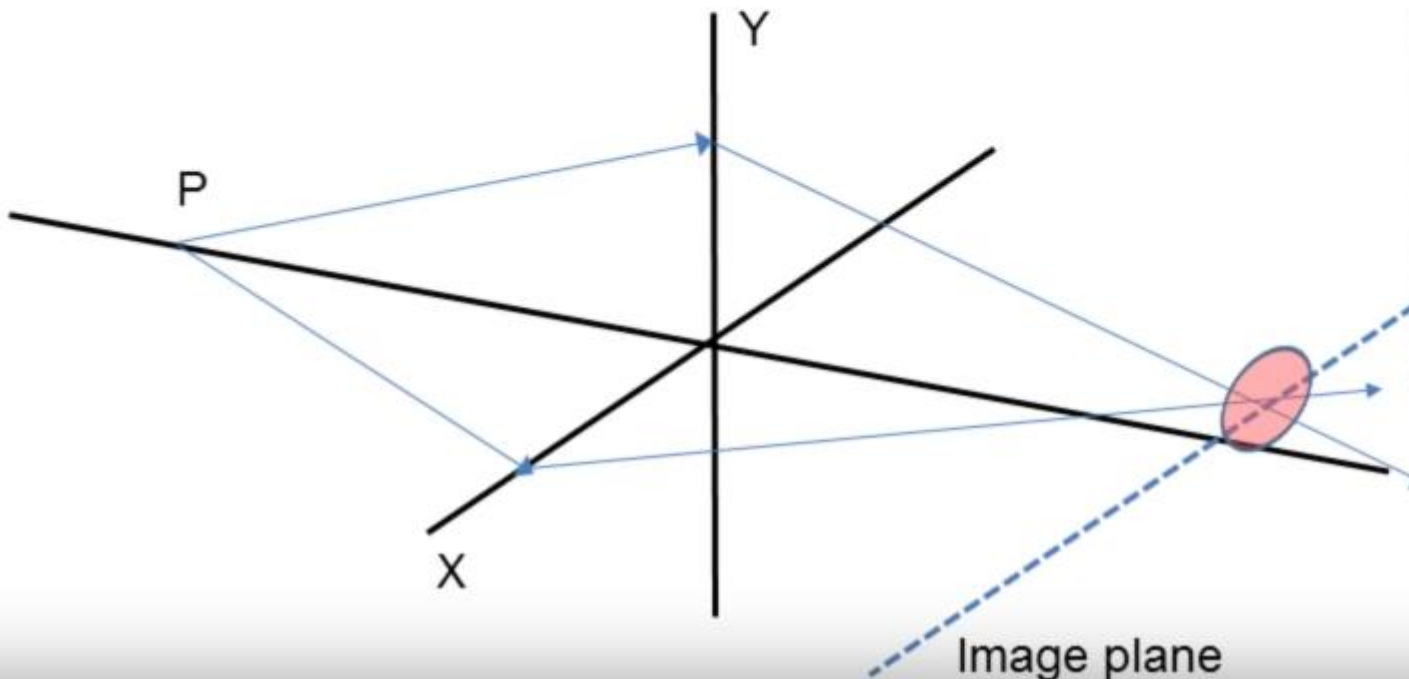
- Electromagnetic lenses perpendicular to the optic axis  
→ Push and pull the beam
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**Quadrupole** lens is basis of a stigmator



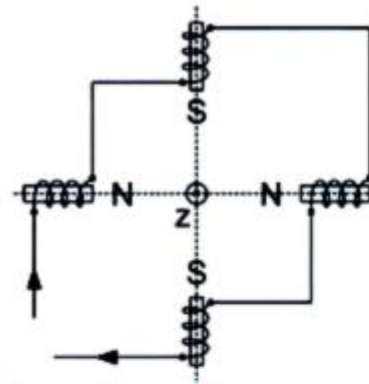
Modern SEMs use **octupole** lens stigmator



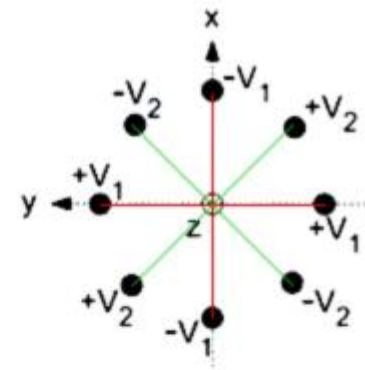
Circular probe:  
Stigmators applied, stretching eliminated

# Shaping the Beam-Deflectors

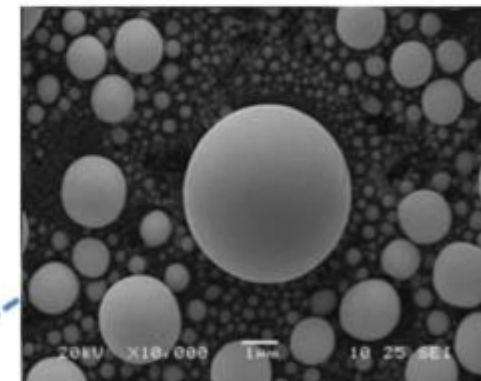
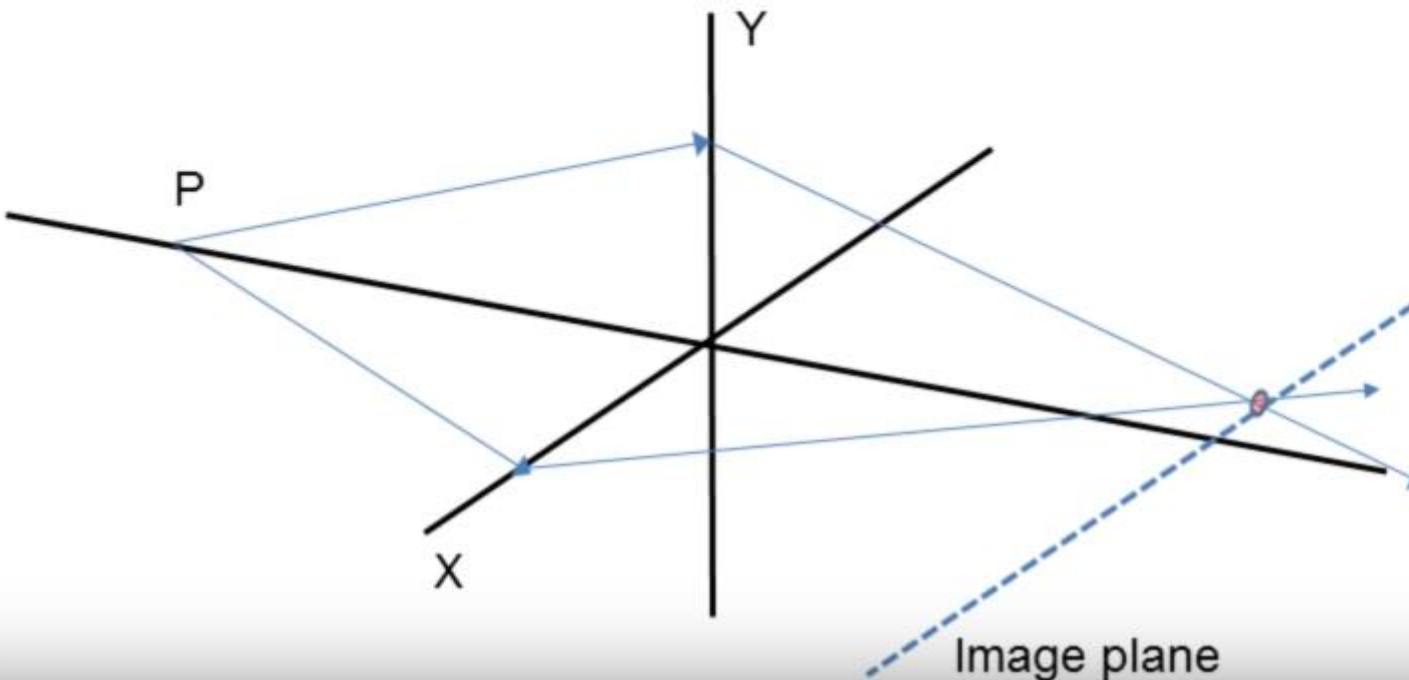
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**Quadrupole** lens is basis of a stigmator



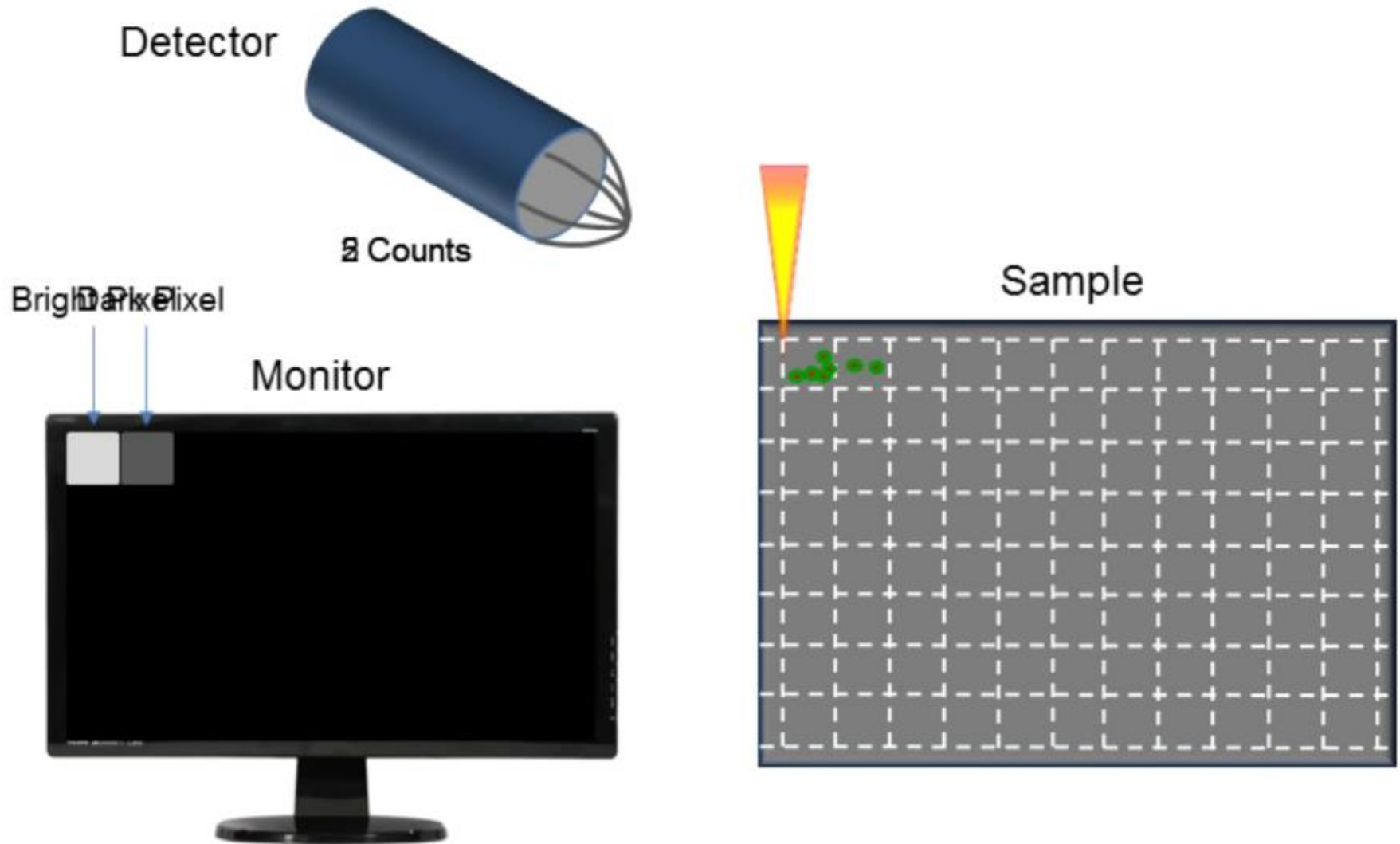
Modern SEMs use **octupole** lens stigmator



Focused probe:  
Corrected  
objective, edges  
sharpened



# Image Formation in SEM

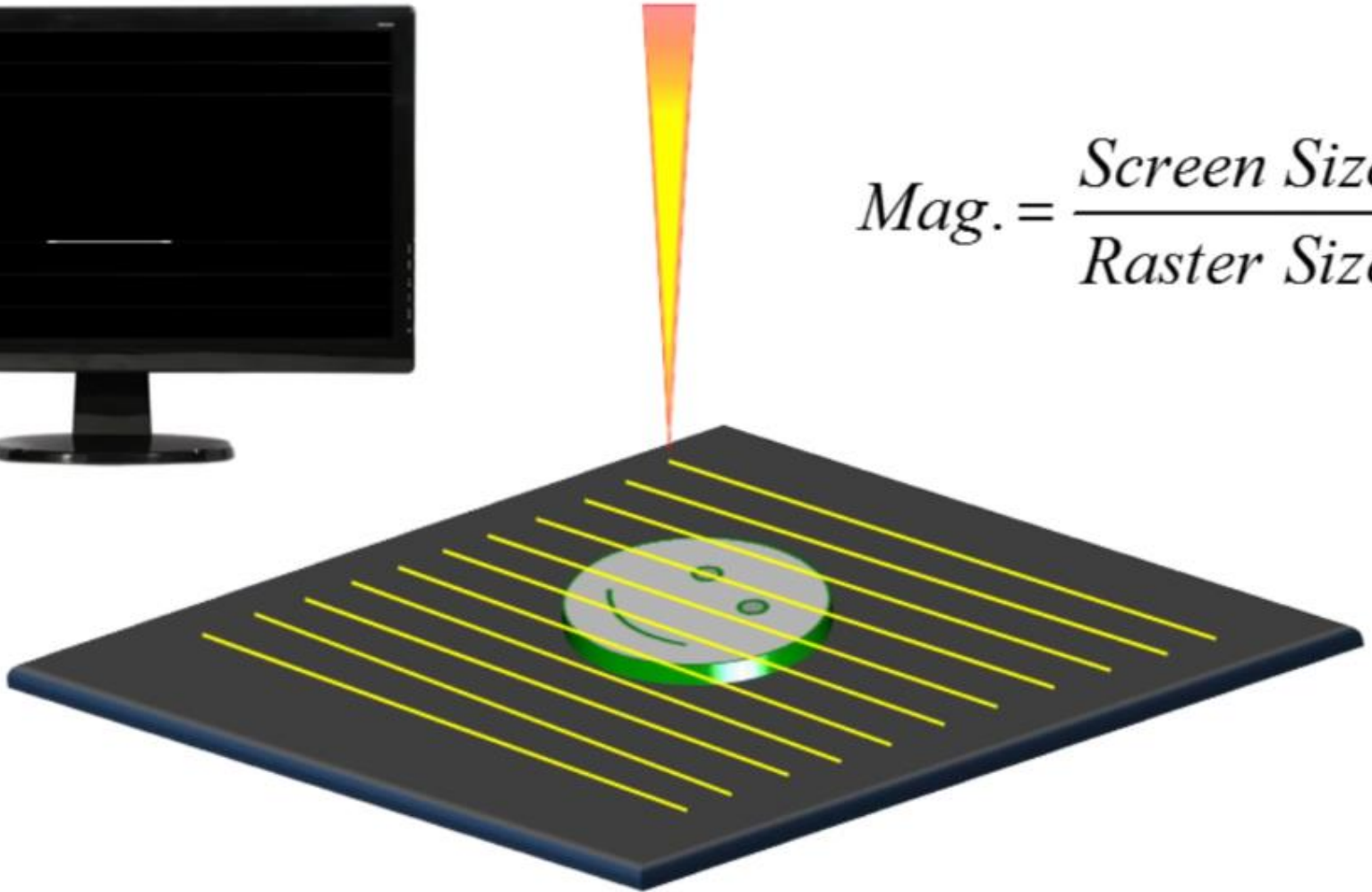


- Pixel intensity is proportional to signal received
- Monitor and scanning (deflection) coils are synchronized

# The “Scanning” part of SEM



$$Mag. = \frac{Screen\ Size}{Raster\ Size}$$



- Deflection coils are used to steer the beam
- Beam continuously scans across the surface of sample, then flies back to starting point

# Quiz

- ◆ What is chromatic aberration, spherical aberration, aperture diffraction and astigmatism, and how they are formed?

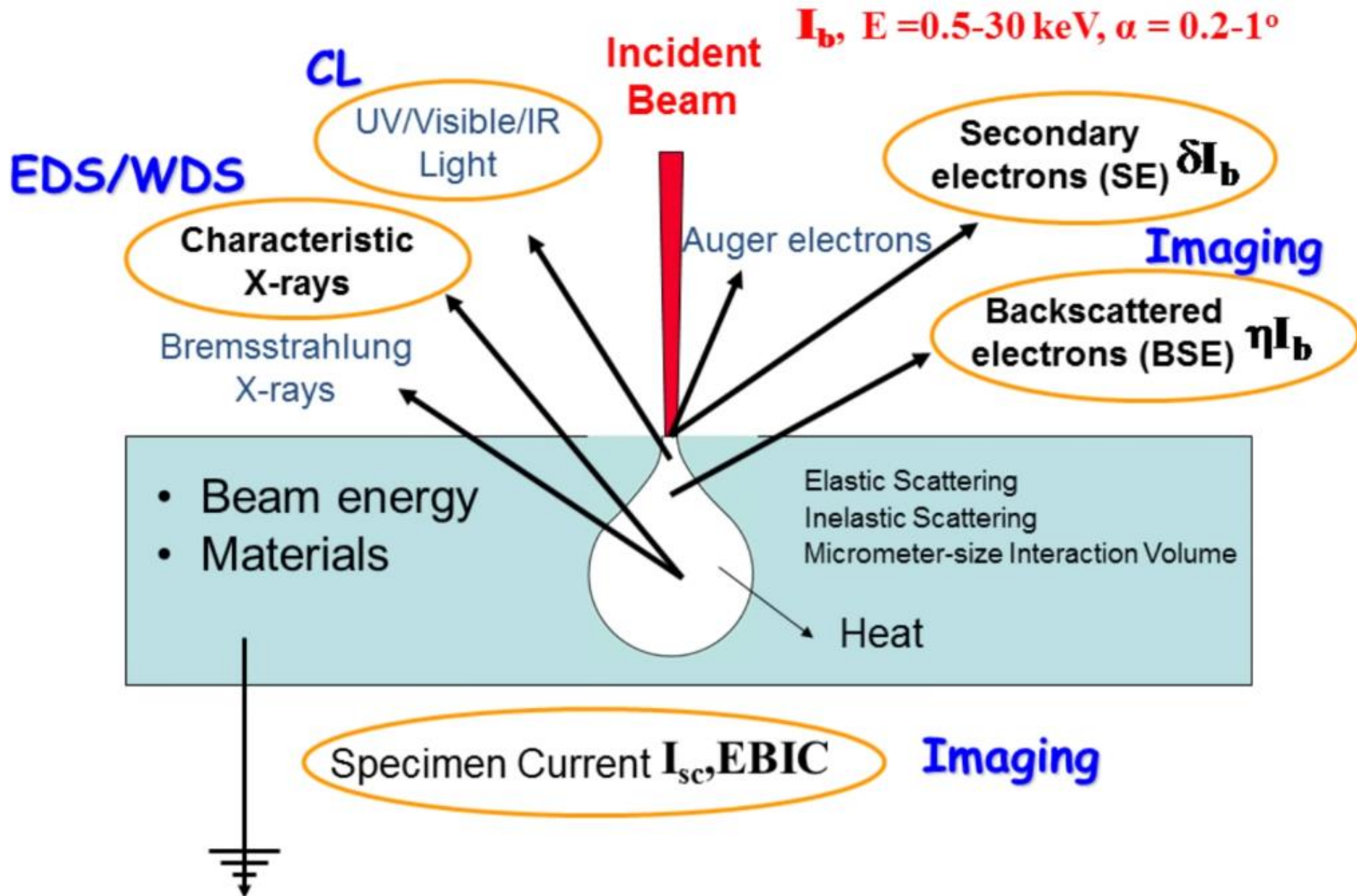


# Electron-Matter Interaction and Detectors





# Electron-Matter Interactions

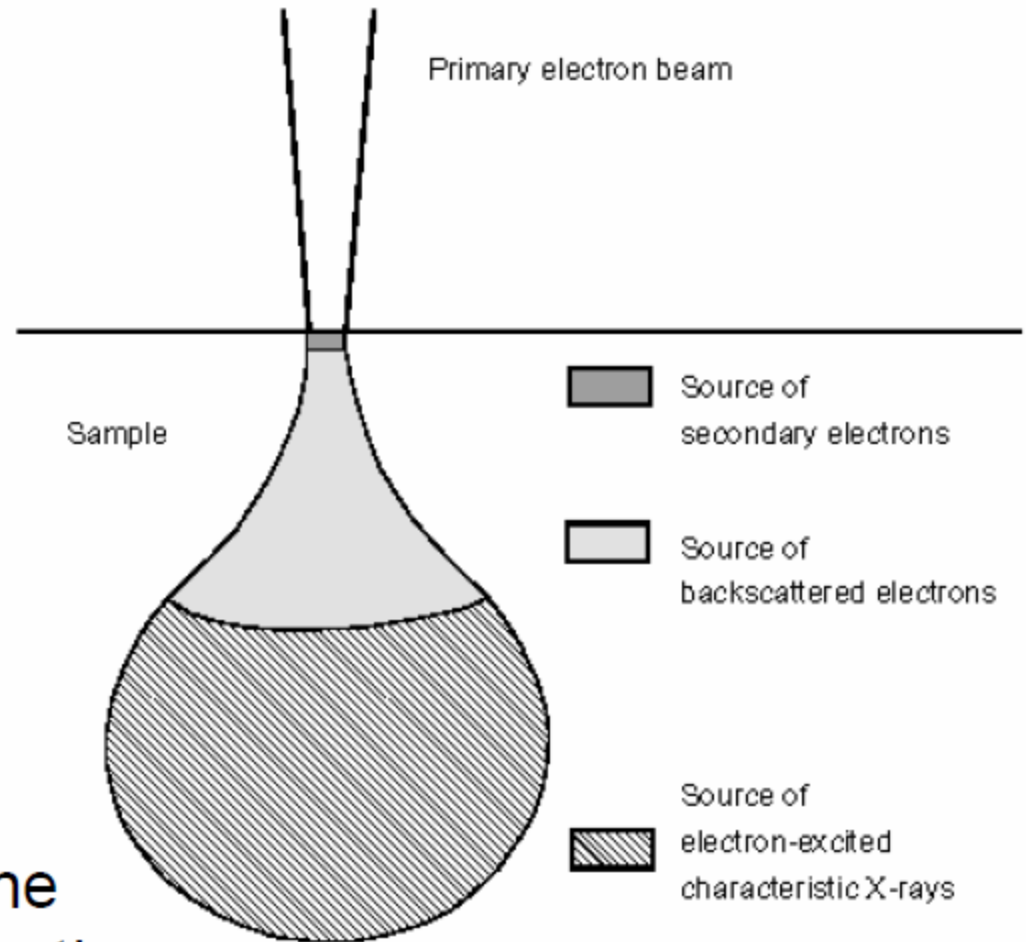


# Effective Excitation Volumes

**Effective** excitation volumes defined by:

- Excitation Energy
- Atomic Number
- Escape Depth

Escape depth often determines the experimental interaction depth (ie. depth resolution).



# Imaging with Secondary Electrons

“SE” are electrons with energies of 10-50 eV produced during ionization of sample atoms.

Emission efficiency is always  $>1$  (i.e. for every incident high energy electron, up to hundreds of low energy secondary electrons are generated).

Depends on:

- **Work function** - composition & orientation.
- **Beam conditions** - # SE's  $\propto$  beam current (but remember current increases with beam spot size<sup>3</sup>).
- **Sample density (Z)** – but usually masked by contamination.
- **Topography** – SE emission enhanced at protrusions.  
***Thus superb for rough samples.***

# SE electron collection

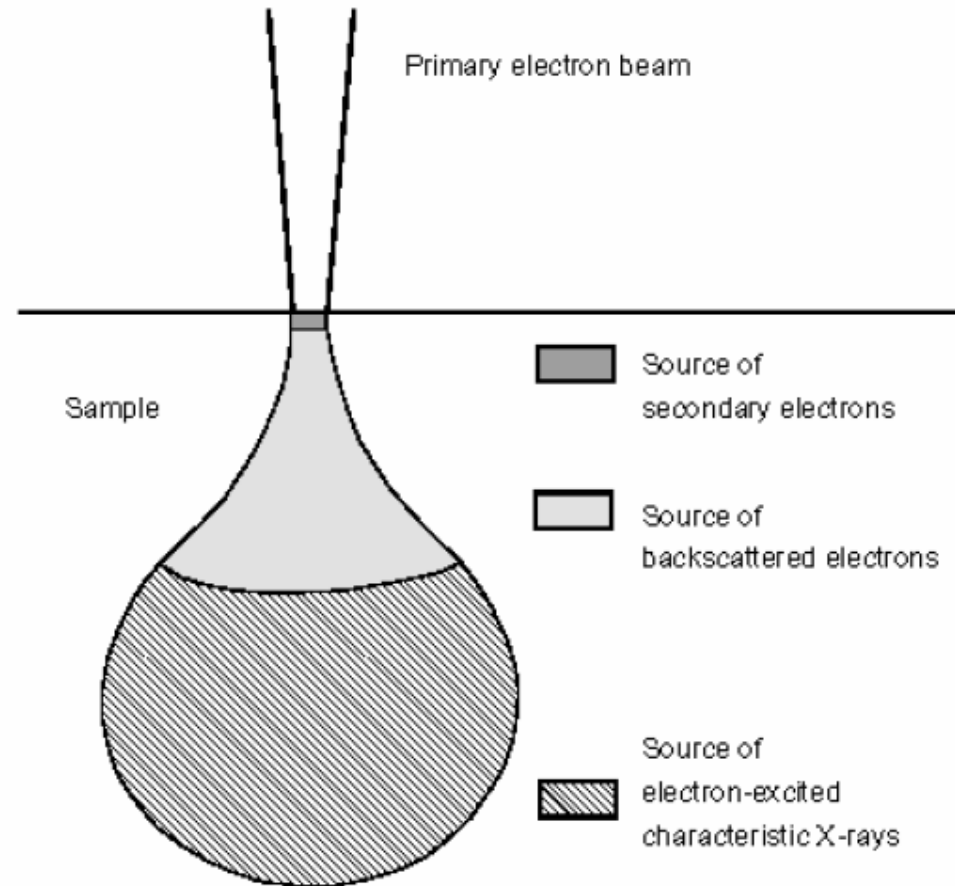
- A detector is placed adjacent to the sample region being imaged (not line of sight).
- The detector is biased with a *small* positive potential, attracting any secondary electrons ejected from the sample.
- Since their energy is low ( $\sim 50$  eV,  $\sim 2$  times  $kT$ , the ave. electron energy at room temperature), mean free paths are low (2 nm).
- Therefore, only secondary electrons generated near the surface escape for detection.
- Although many, many more secondary electrons are generated deep within the sample, they do not contribute to SE imaging contrast. ***Primarily surface sensitive.***
  - For this reason, as incident beam energy increases, # SE's counter-intuitively decreases since most SE's generated are too far from the surface to escape and be detected.



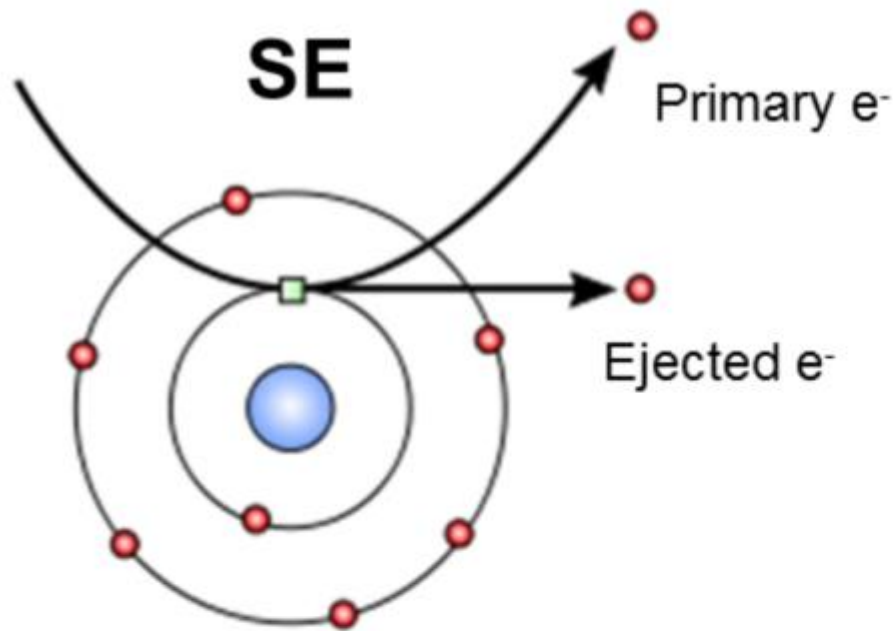


# SE lateral resolution

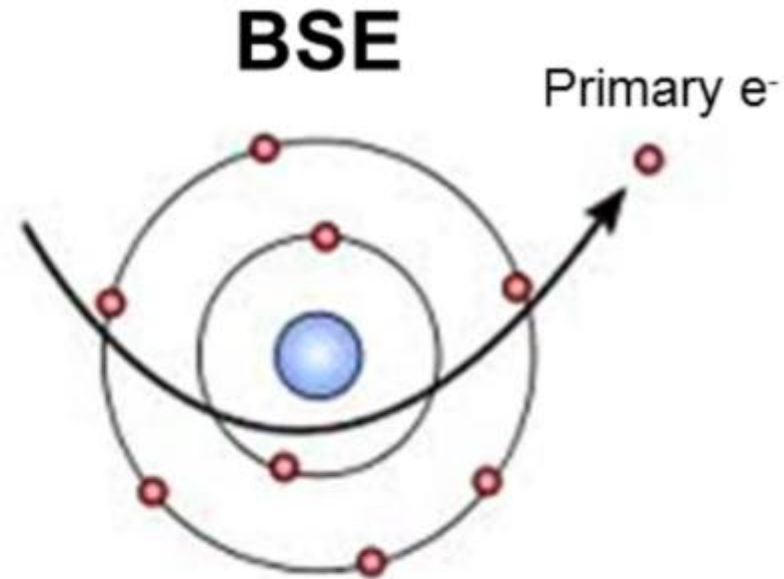
- Radius of the teardrop shape typical for electron-sample interactions is many times larger than the beam spot size (100 times larger or more).
- Secondary electrons from this large area are generated sufficiently below the surface that they do not modify lateral resolution.
- ***Lateral Resolution depends on beam focus, not interaction volume.***



# Generation of Secondary and Backscatter Electrons

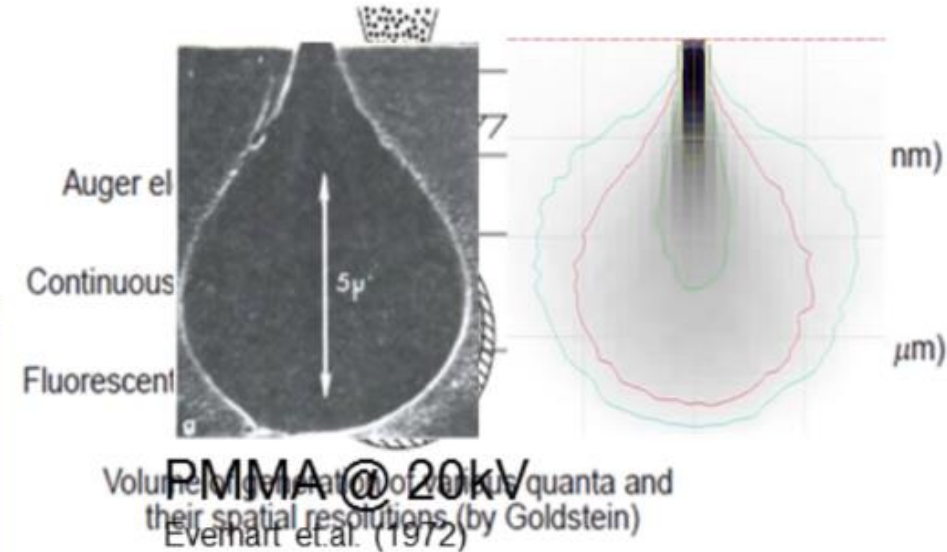
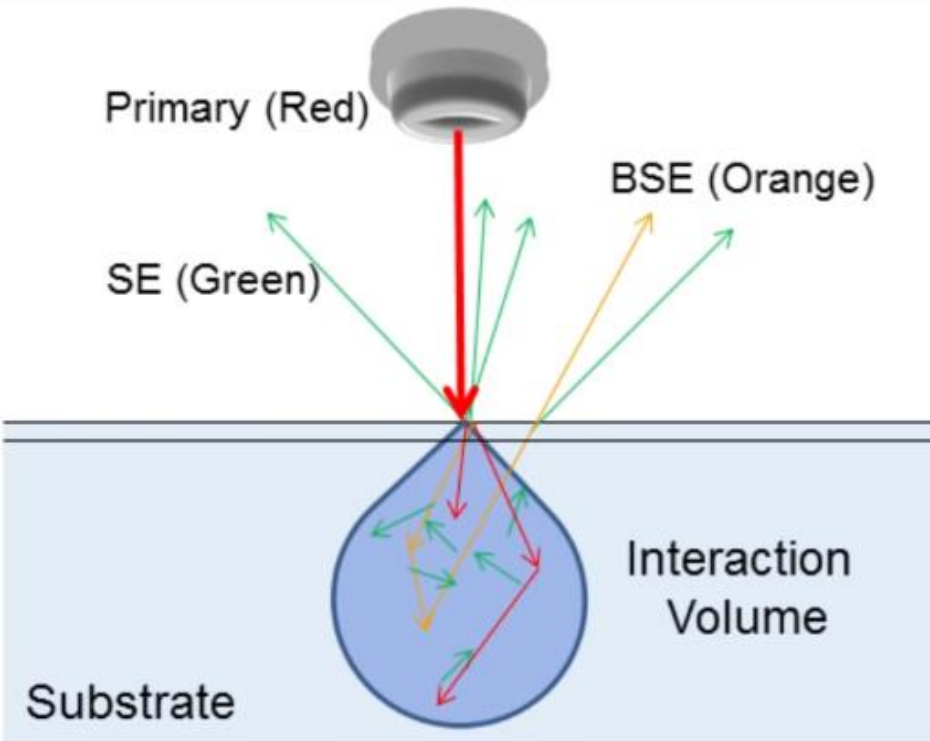


**Secondary electrons**  
are low energy  
electrons ( $< 50\text{eV}$ )  
ejected from the  
specimen atoms by  
the energetic primary  
beam



**Backscattered electrons**  
are primary beam  
electrons scattered  
back out of the sample.  
BSE have a range of  
energies (up to the  
primary energy)

# Interaction Volume of Primary Electrons

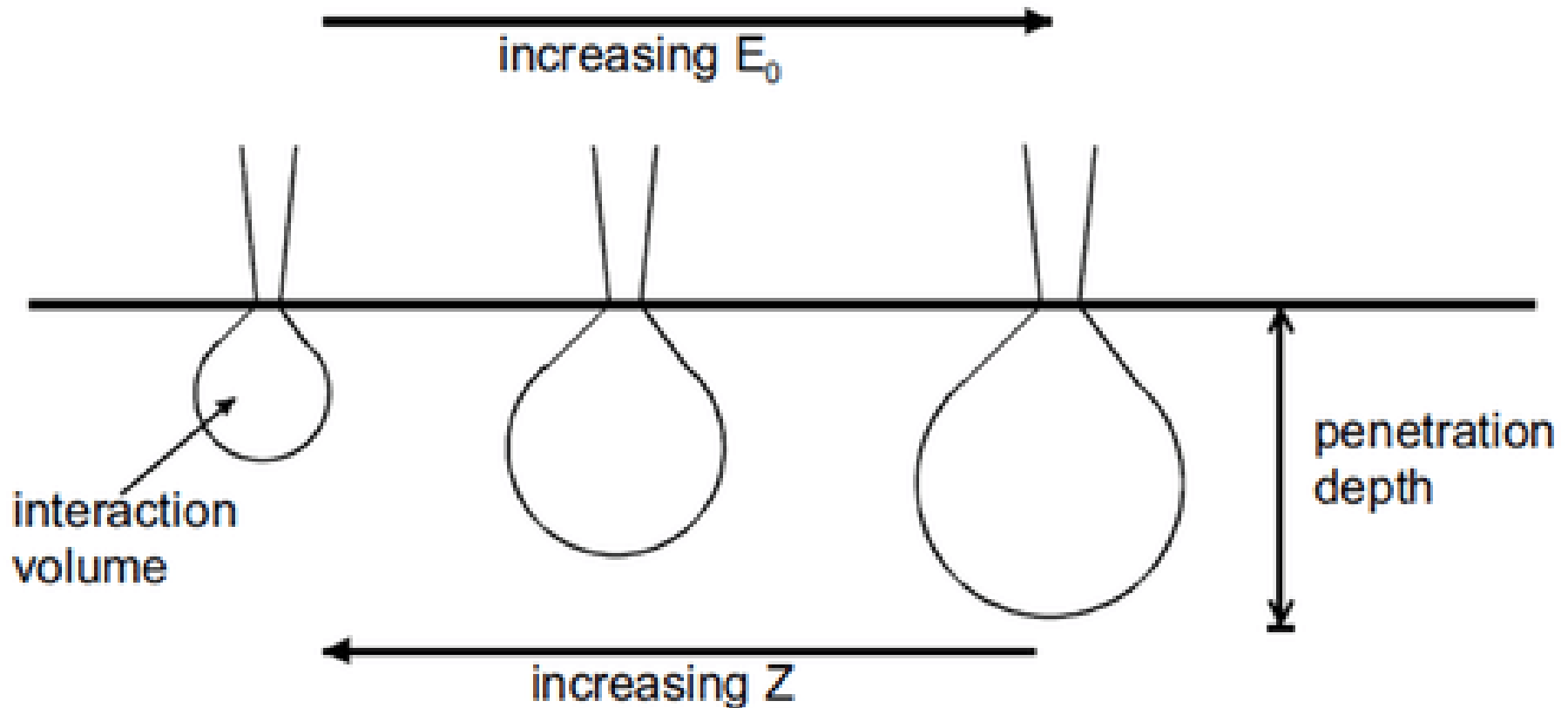


| Type  | Energy | Range          | Sensitivity        |
|-------|--------|----------------|--------------------|
| SE    | <50eV  | Few nm (5-10)  | Surface topography |
| BSE   | <PE    | Partial volume | Material (Ave. Z)  |
| X-ray | <PE    | Full volume+   | Element            |

Higher primary energy → Larger interaction volume  
Keep the interaction volume in mind when analyzing samples!

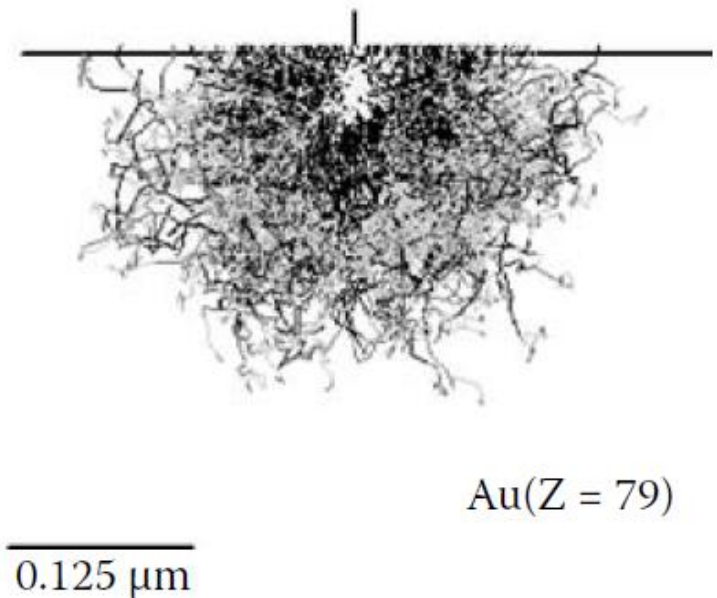
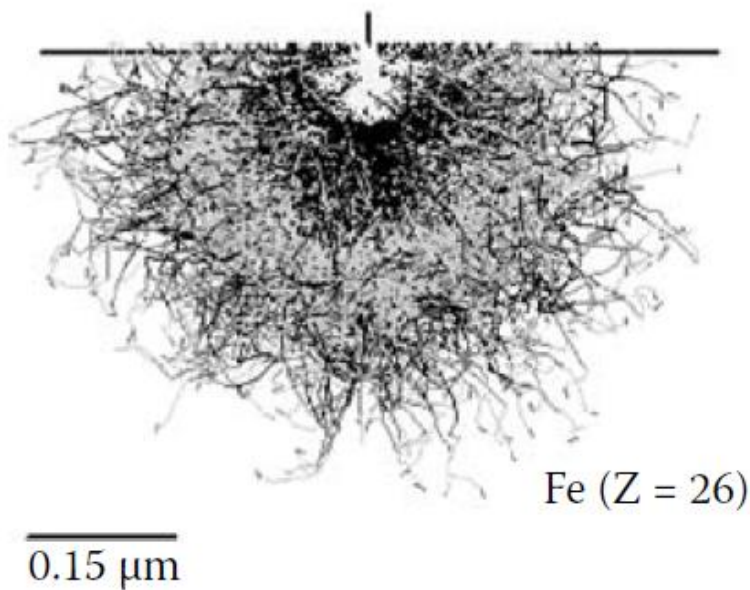
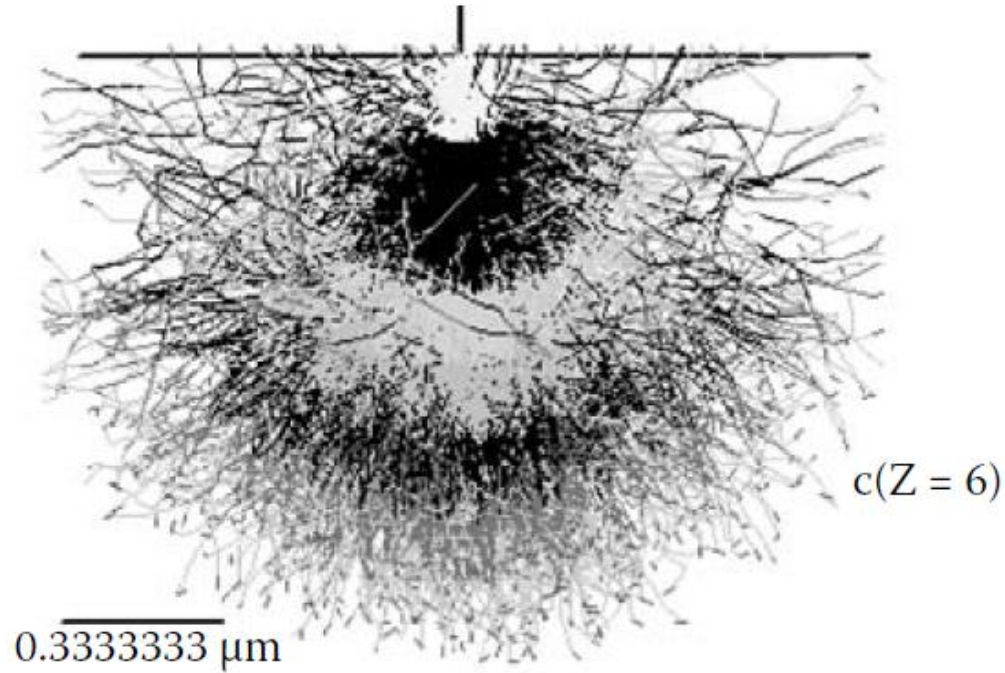
# Interaction Volume of Primary Electrons

- Interaction volume increased with  $E$
- Interaction volume decreased with  $Z$

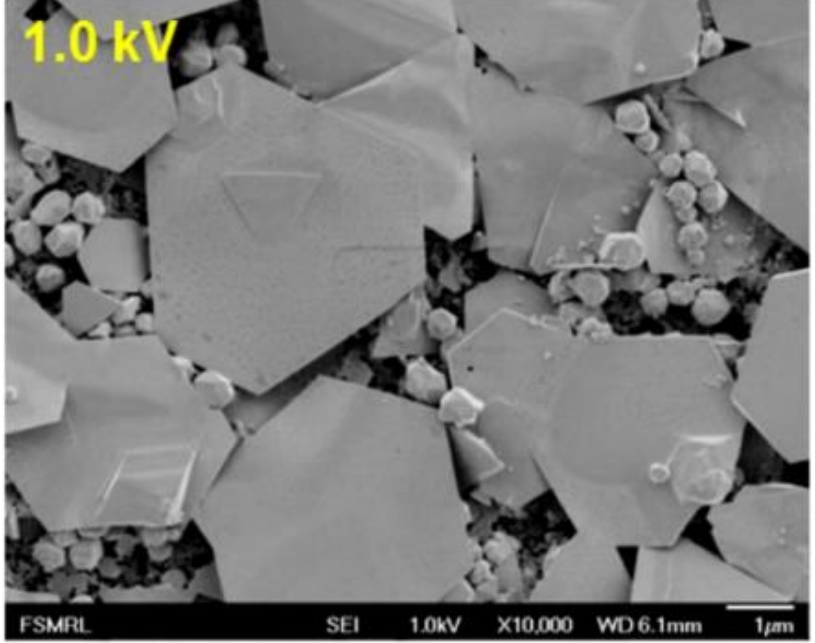
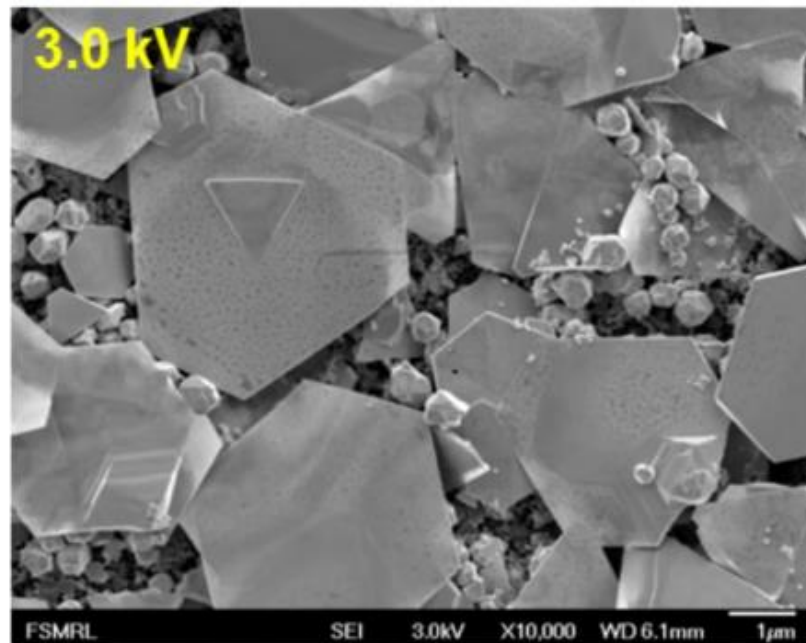
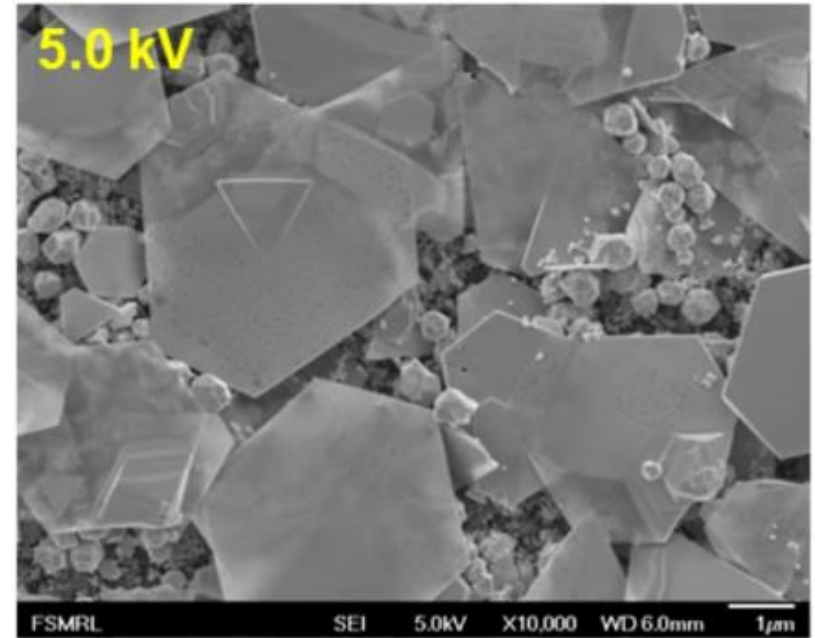
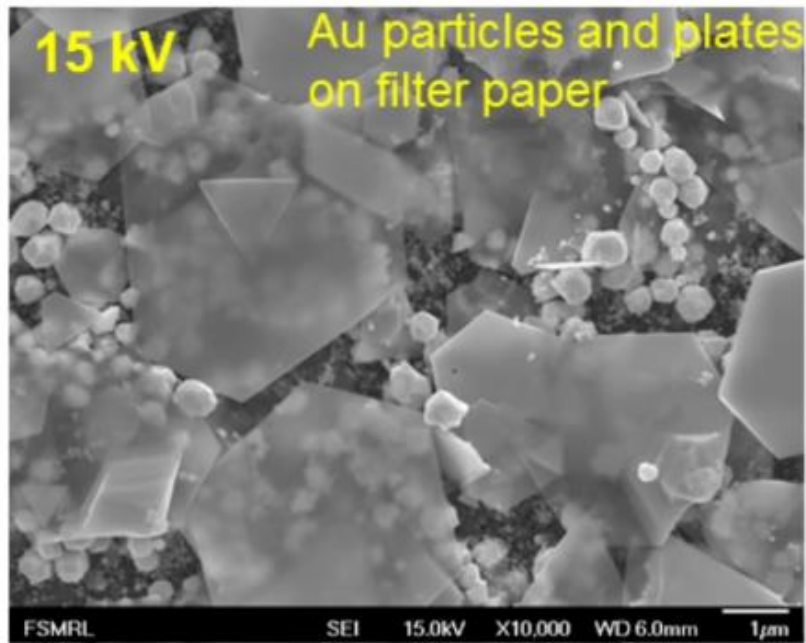




# Interaction Volume of Primary Electrons: different specimen

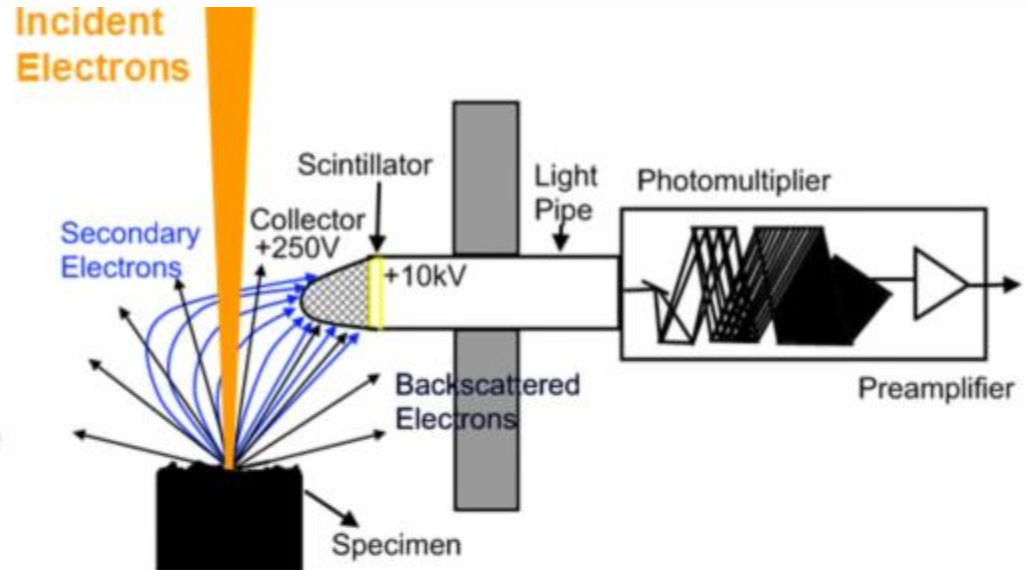


# Beam energy and SE Imaging

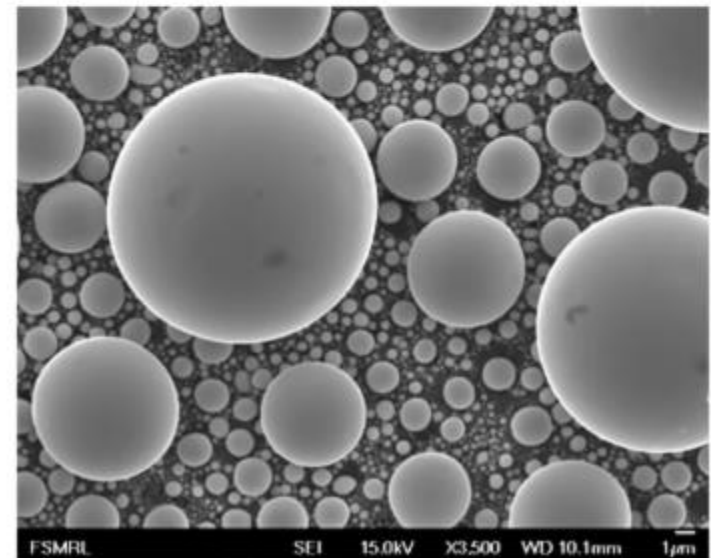
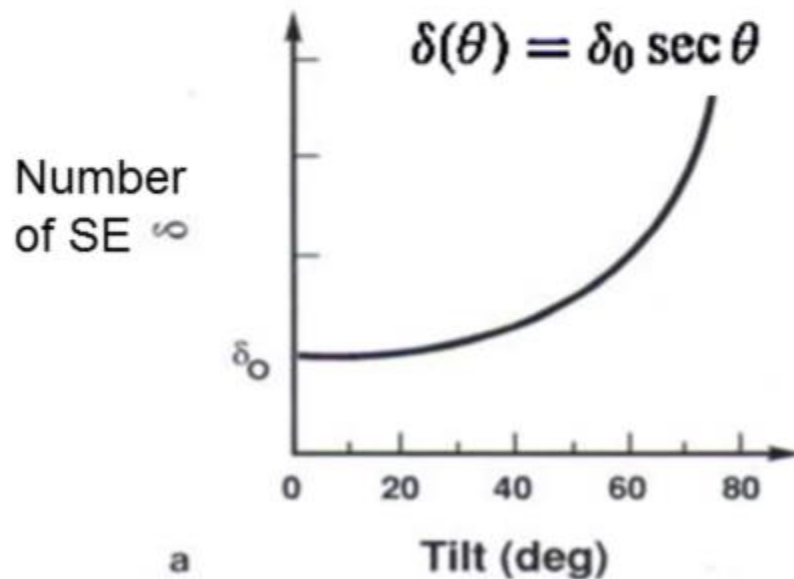


# SE Detector and Imaging: Topography

- Biased grid attracts low energy SE
- Collects some BSE from solid-angle
- SE emission from surface sensitive to sample curvature



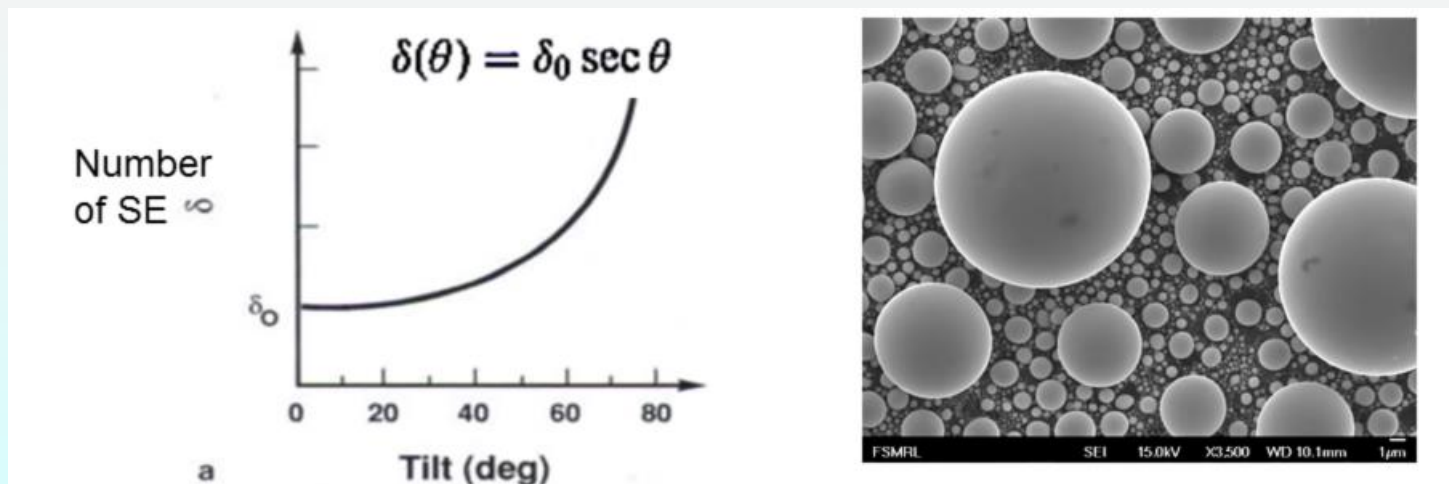
Everhart-Thornley SE detector





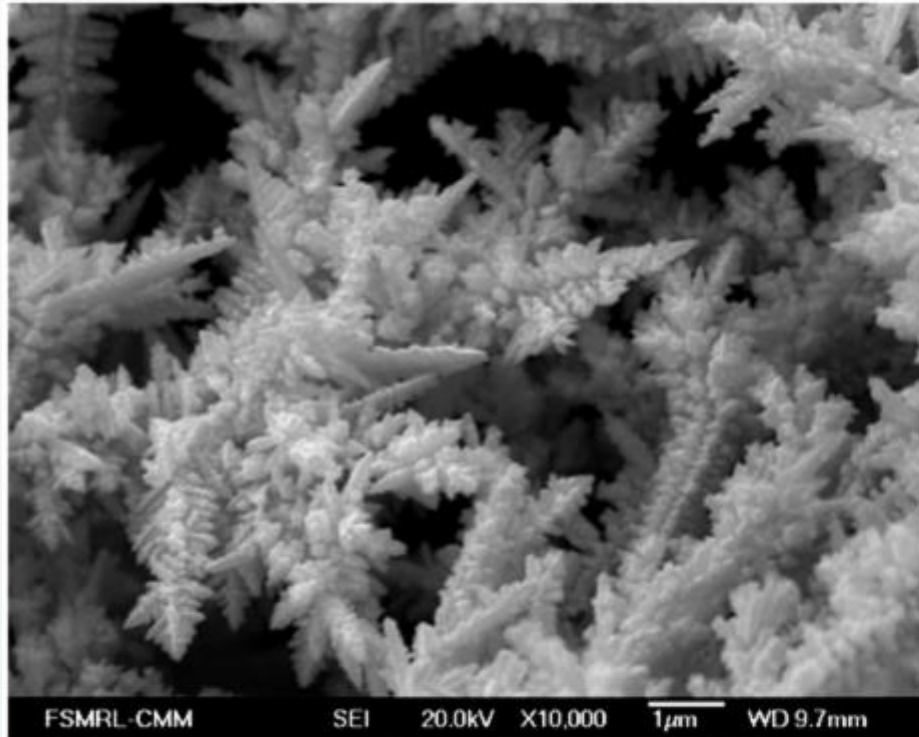
# Homework 4

1. For the electron-matter interactions, what signals will be generated and how they are formed? What they can be used for?
2. The more SE are collected at edges, explain

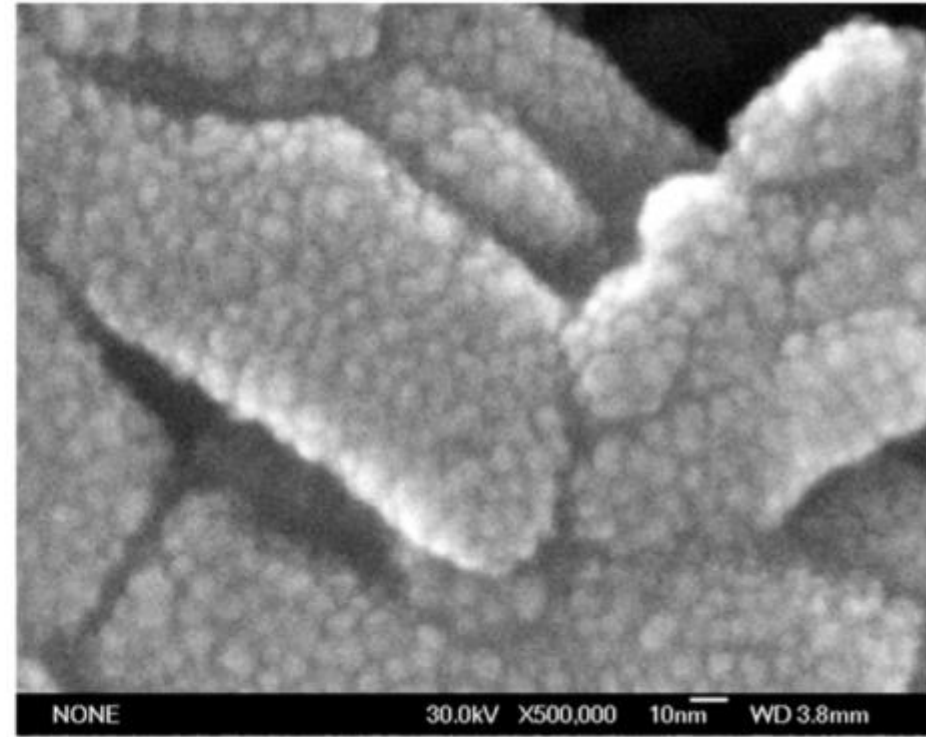




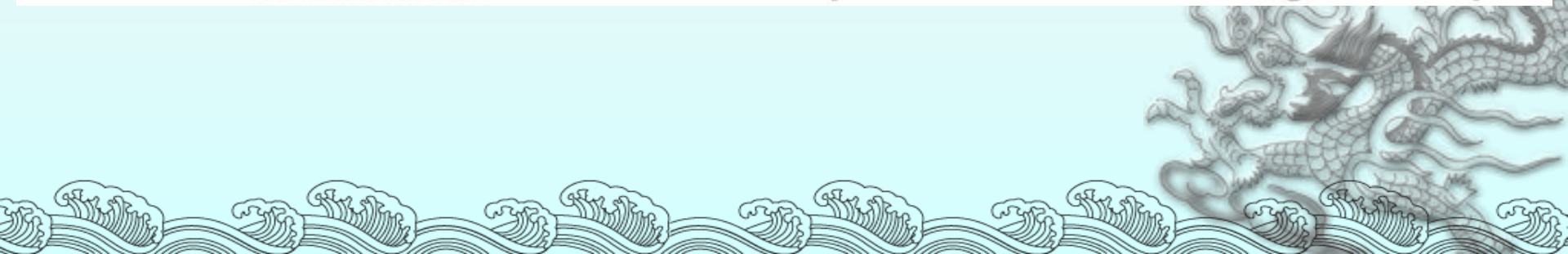
# Examples of SE Imaging: Topography



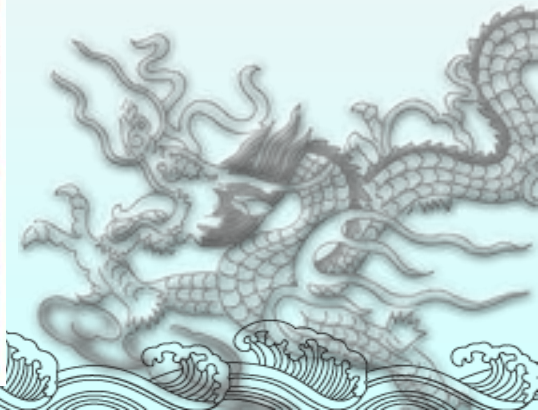
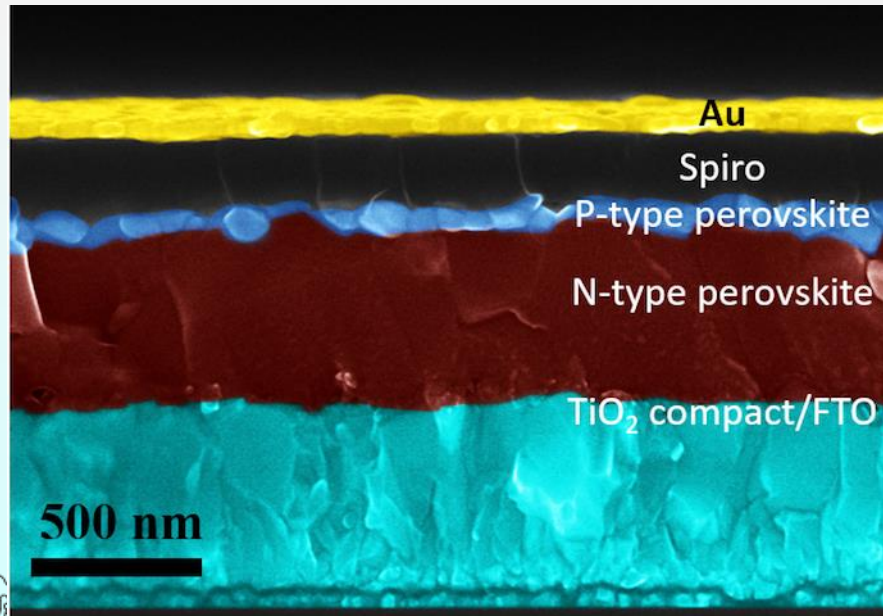
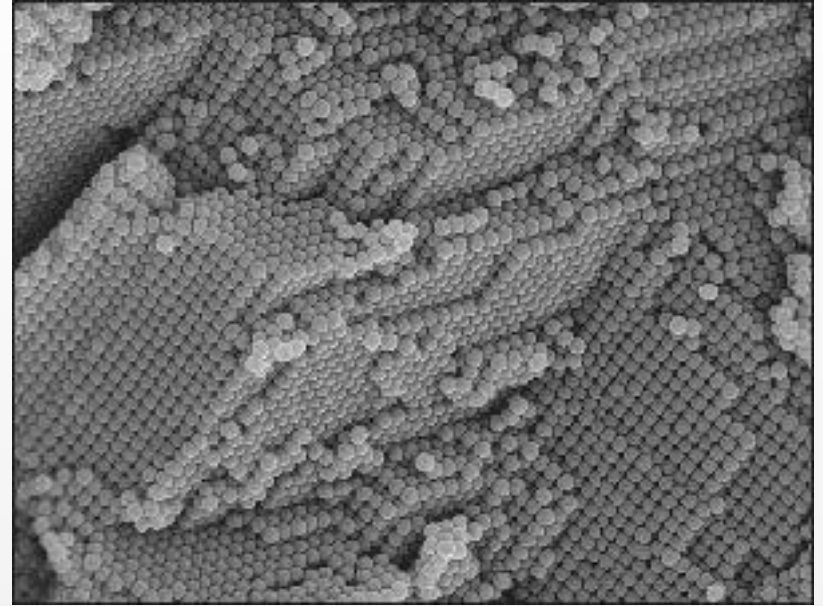
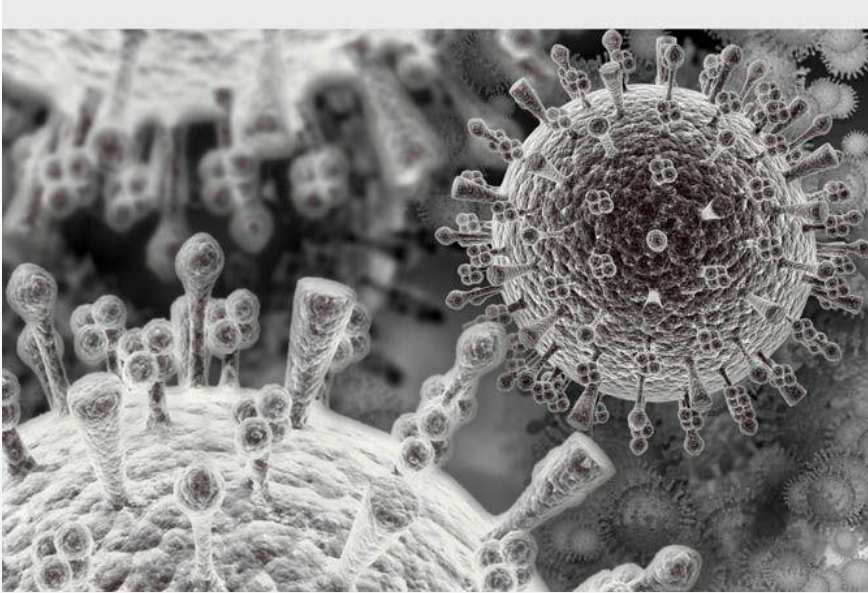
**Dendritic Au**



**Sputtered Au-Pd on Magnetic Tape**

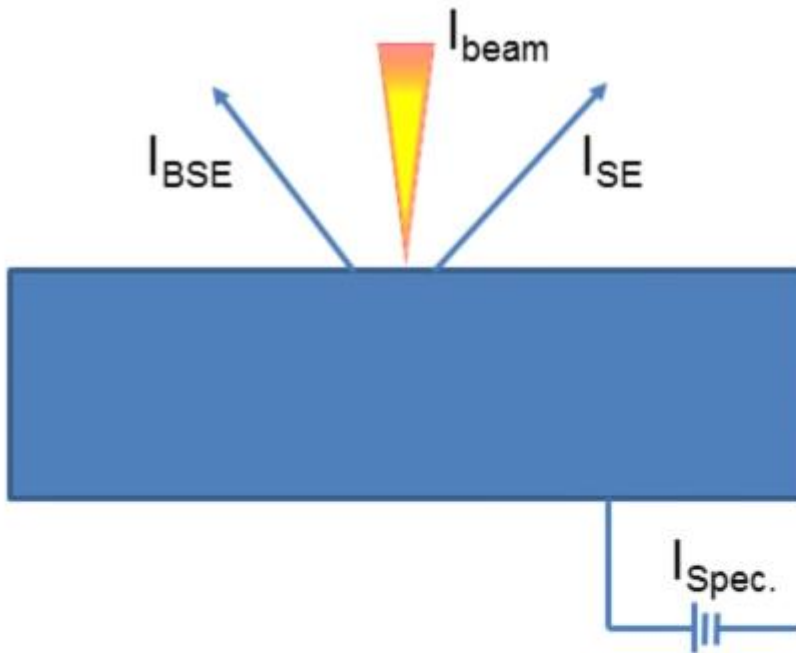


# Observe the Nanoscale World!

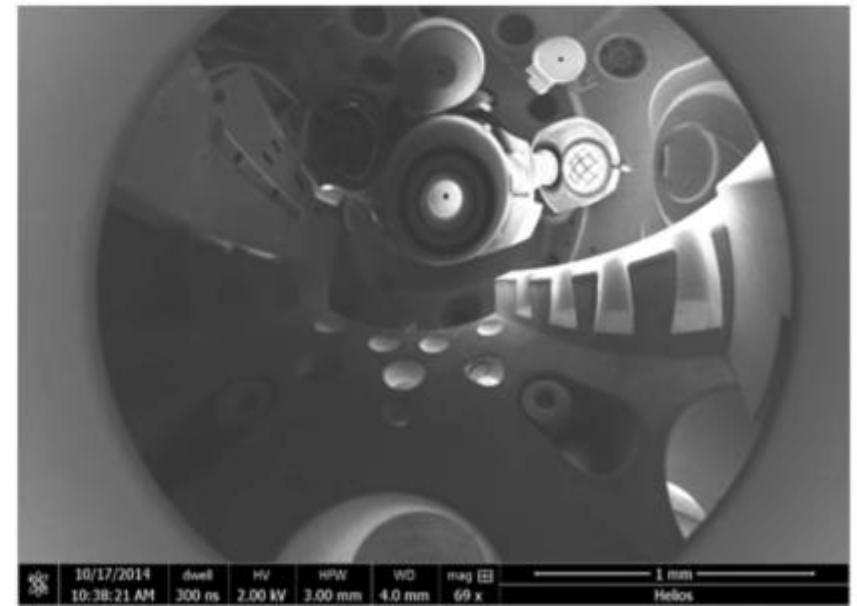
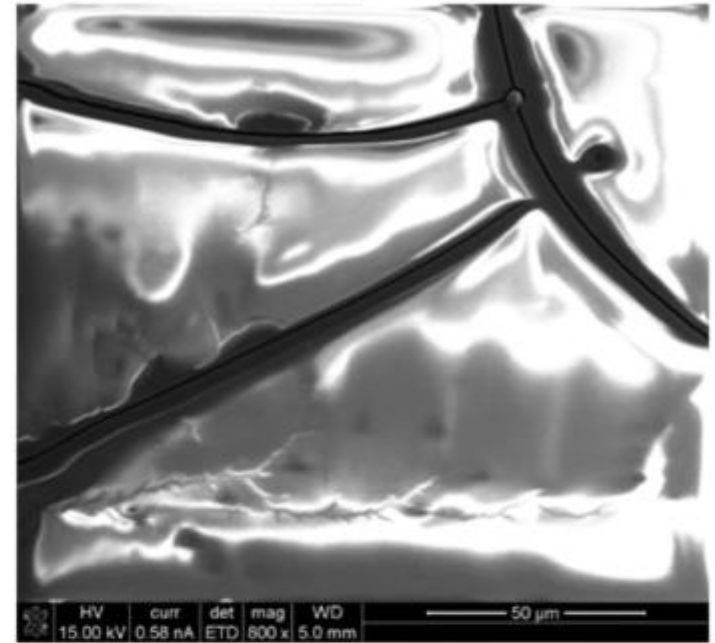




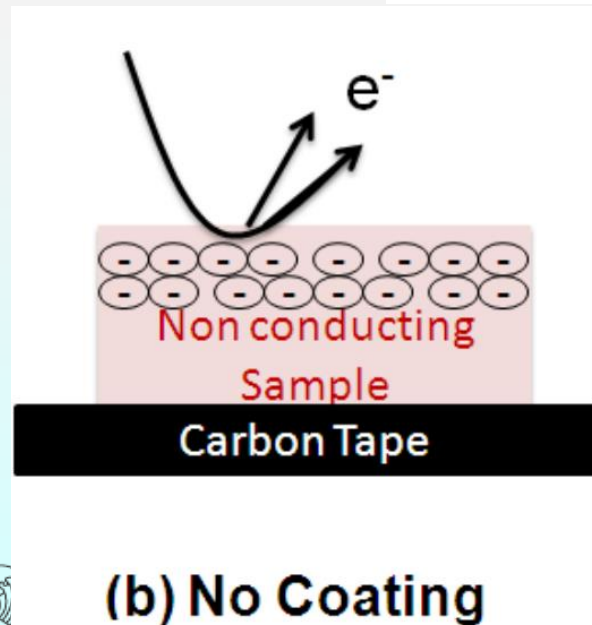
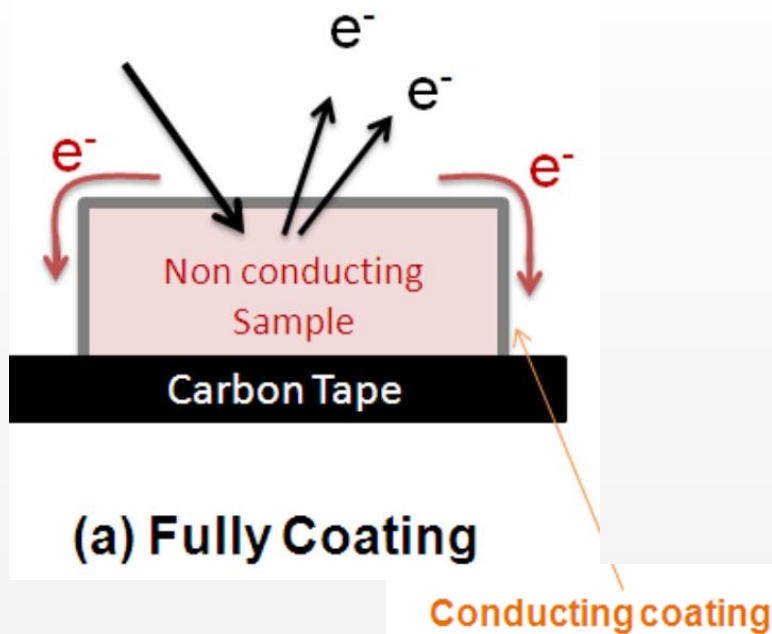
# Electrons Inject Charge-Where do they go?



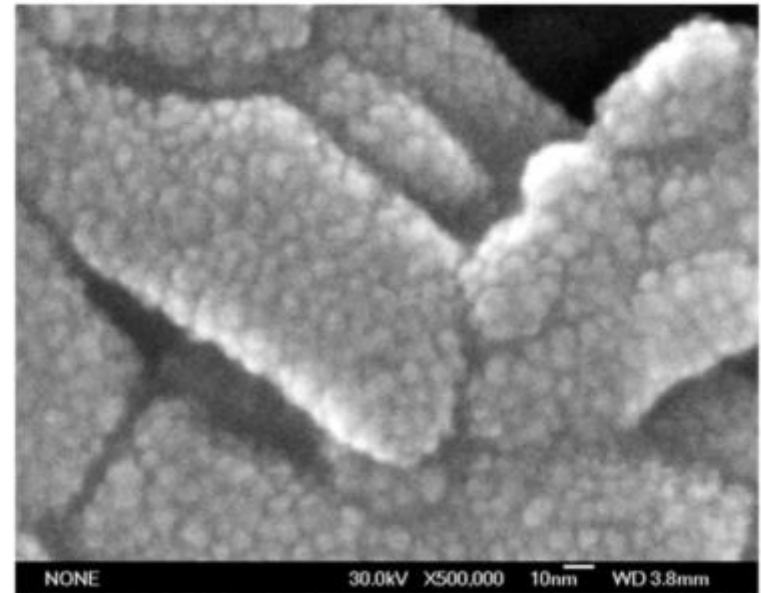
- If the sample is conducting there is no charge buildup
- If the sample is insulating the electron beam will charge the sample surface – image will be distorted!



# Non-conductive Specimens-Coating and Beam



**The traditional solution**  
- coat with few nm of conductor,  
(e.g. Au/Pd or C) to make sample  
conductive

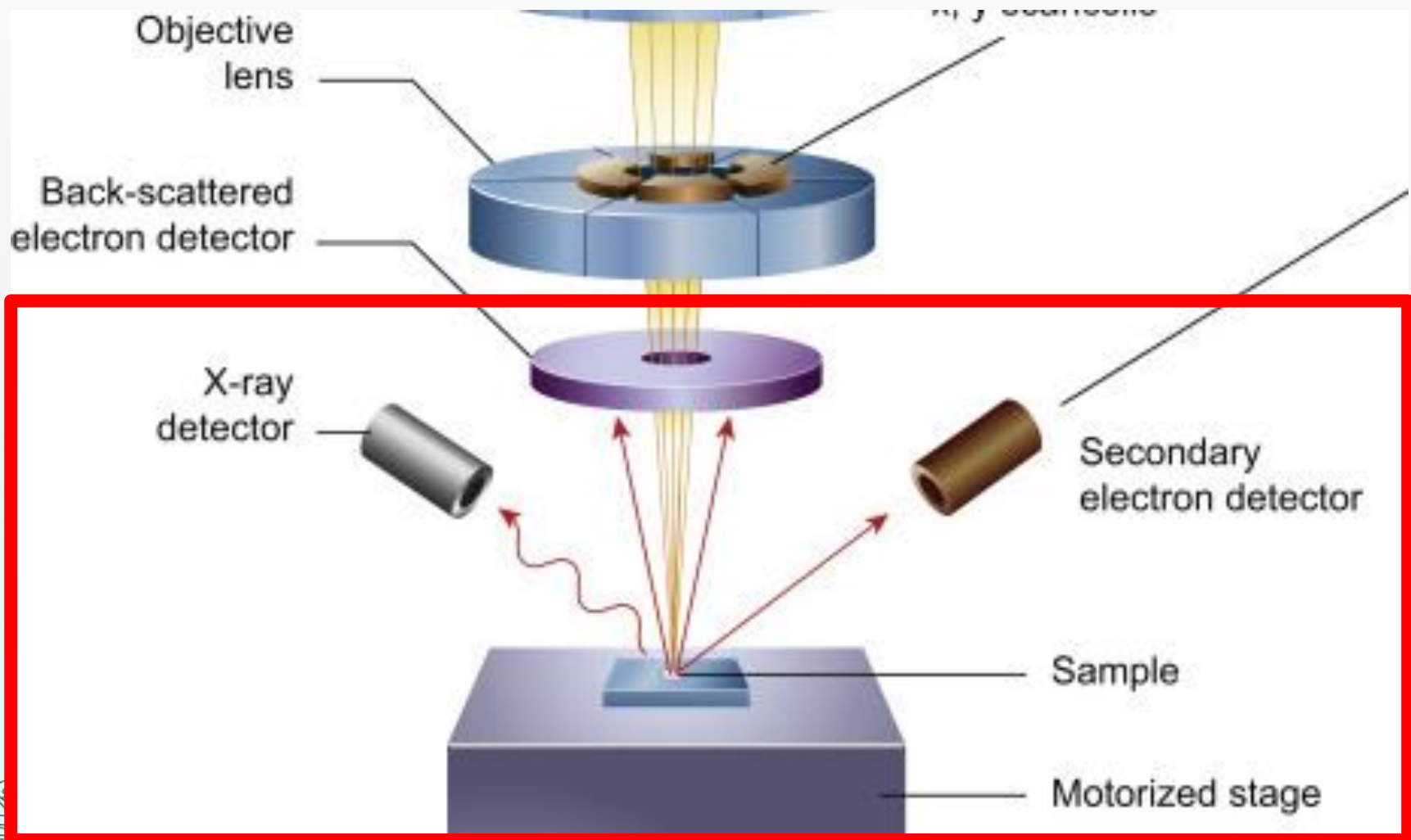


**Sputtered Au-Pd on Magnetic Tape**



# Specimen Chamber

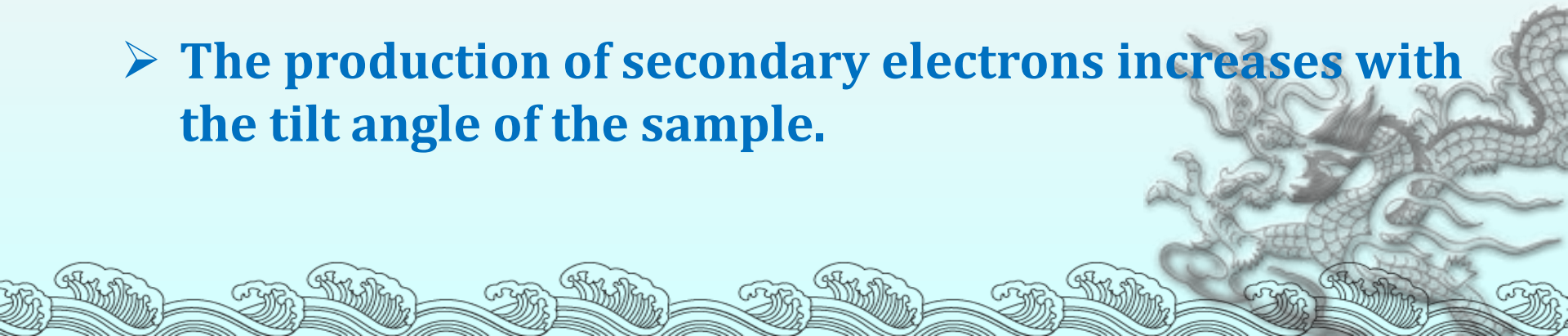
- Vacuum in the chamber
- Sample should be solid and dry



# Remarks for Secondary Electrons

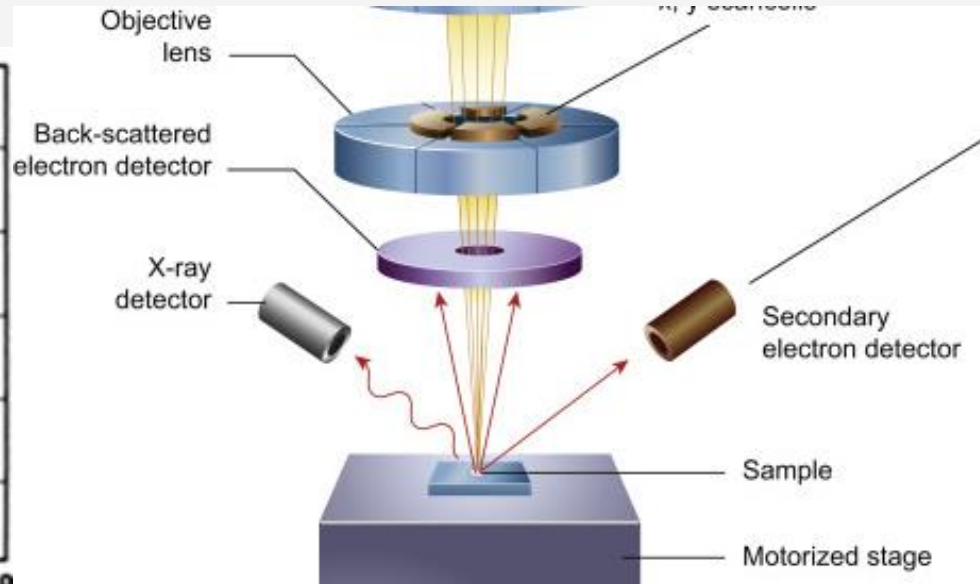
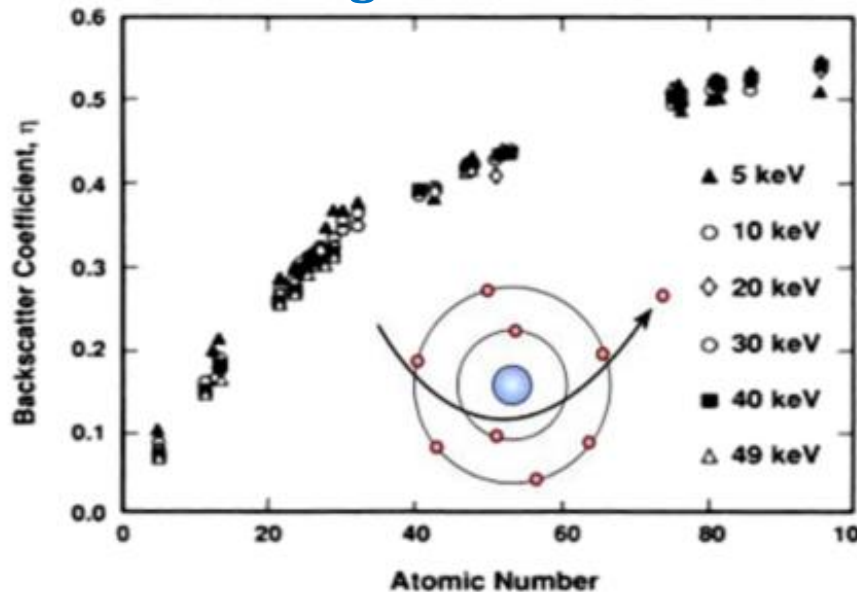
## Some of the properties of secondary electrons

- Secondary electrons are highly sensitive to topographical features of the specimen surface because they depend on the surface orientation with respect to the electron beam.
- There is no significant dependence between the atomic number of the specimen and the intensity of the secondary electron signal.
- The production of secondary electrons increases with the tilt angle of the sample.



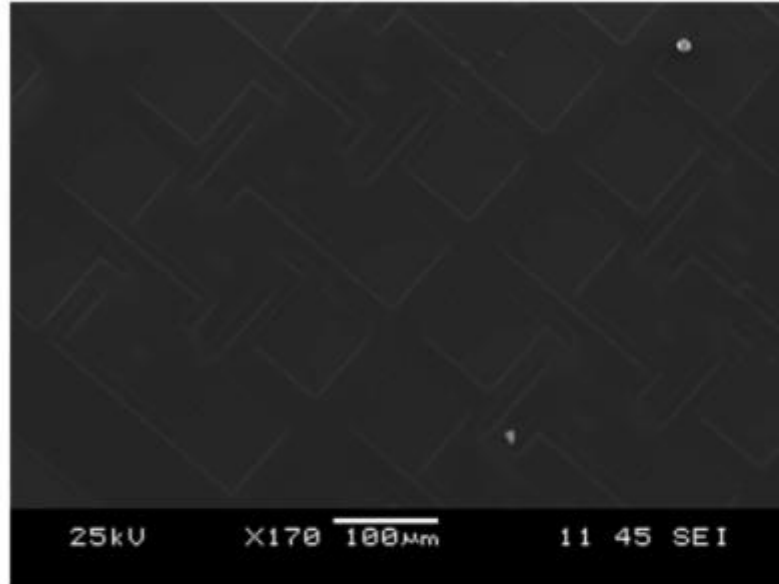
# Composition Contrast with BSE Detector

- Scattering increase with sample Z
- Quadrant detector placed underneath pole-piece
- Compositional mode imaging most useful on multi-phase samples
- Sensitivity can be as low as 0.1 average Z difference

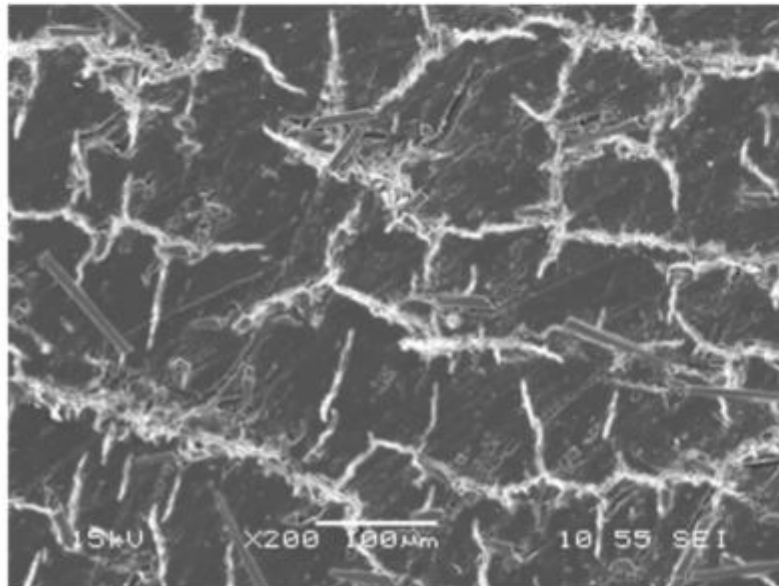


# BSE: Material contrast

SE



BSE





# Electron Backscatter Diffraction

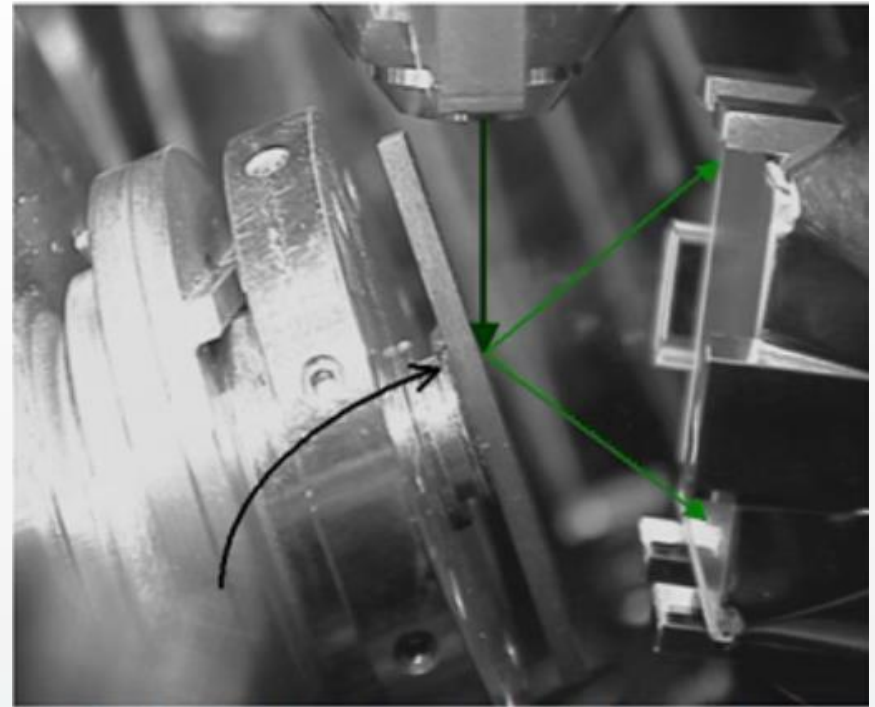
- What is EBSD?
- Why use EBSD?
  - Why measure microstructure?
  - What does EBSD do that can't already be done?
- What are the metrological challenges?
- Examples of applications



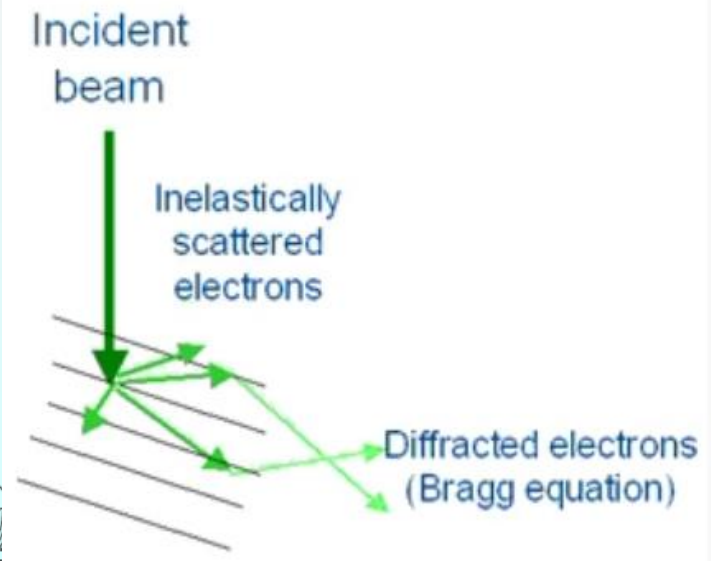
EBSD is a SEM–based microstructural-crystallographic characterization technique commonly used in the study of **crystalline or polycrystalline materials**. The technique can provide information about the **structure, crystal orientation, phase, or strain** in the material.



# Electron Backscatter Diffraction

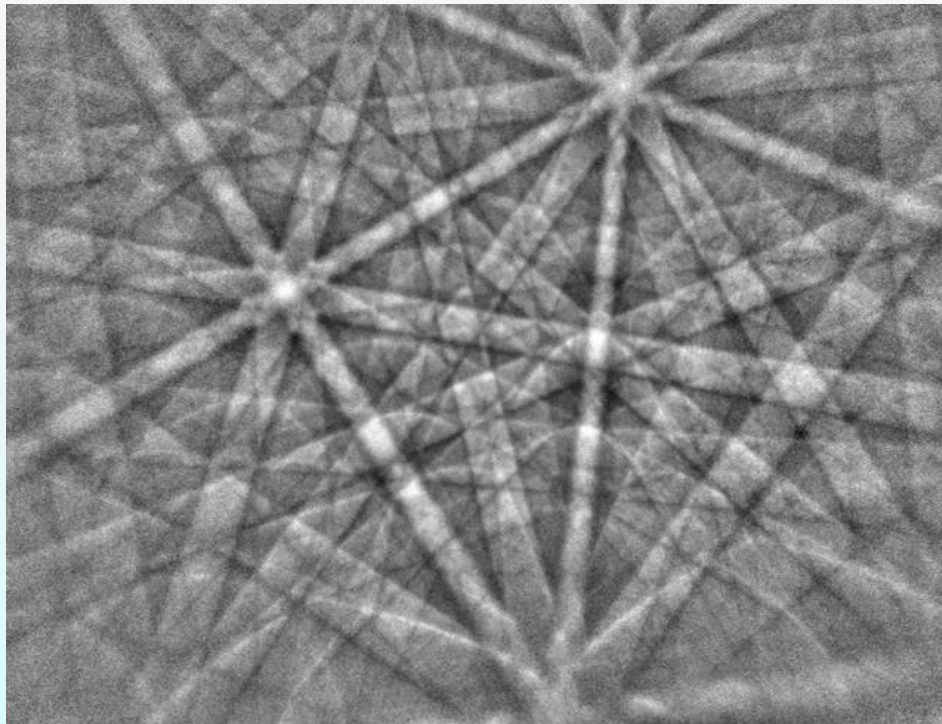


- Prepare sample surface
- Examine in Scanning Electron Microscope (SEM)
- Tilt sample to detect diffracted electrons on phosphor

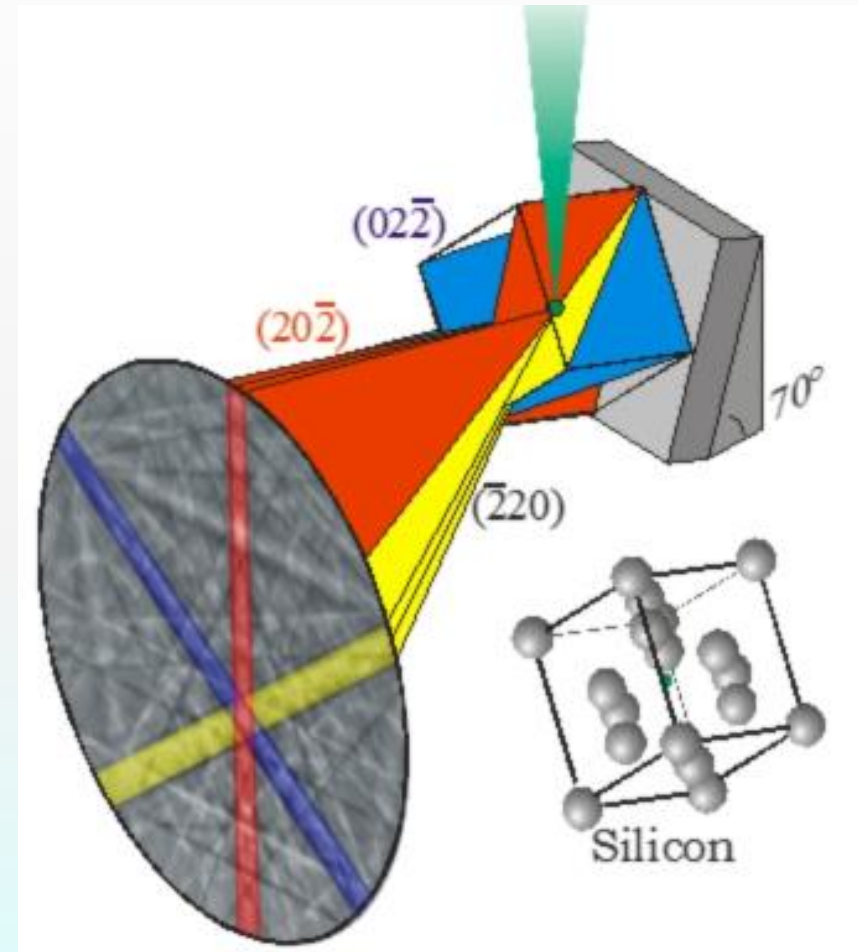


# Electron Backscatter Patterns

- BSE scattered from near-surface form Kikuchi bands
- Gives local orientation and microtexture
- Approximately 50nm (pixel) resolution



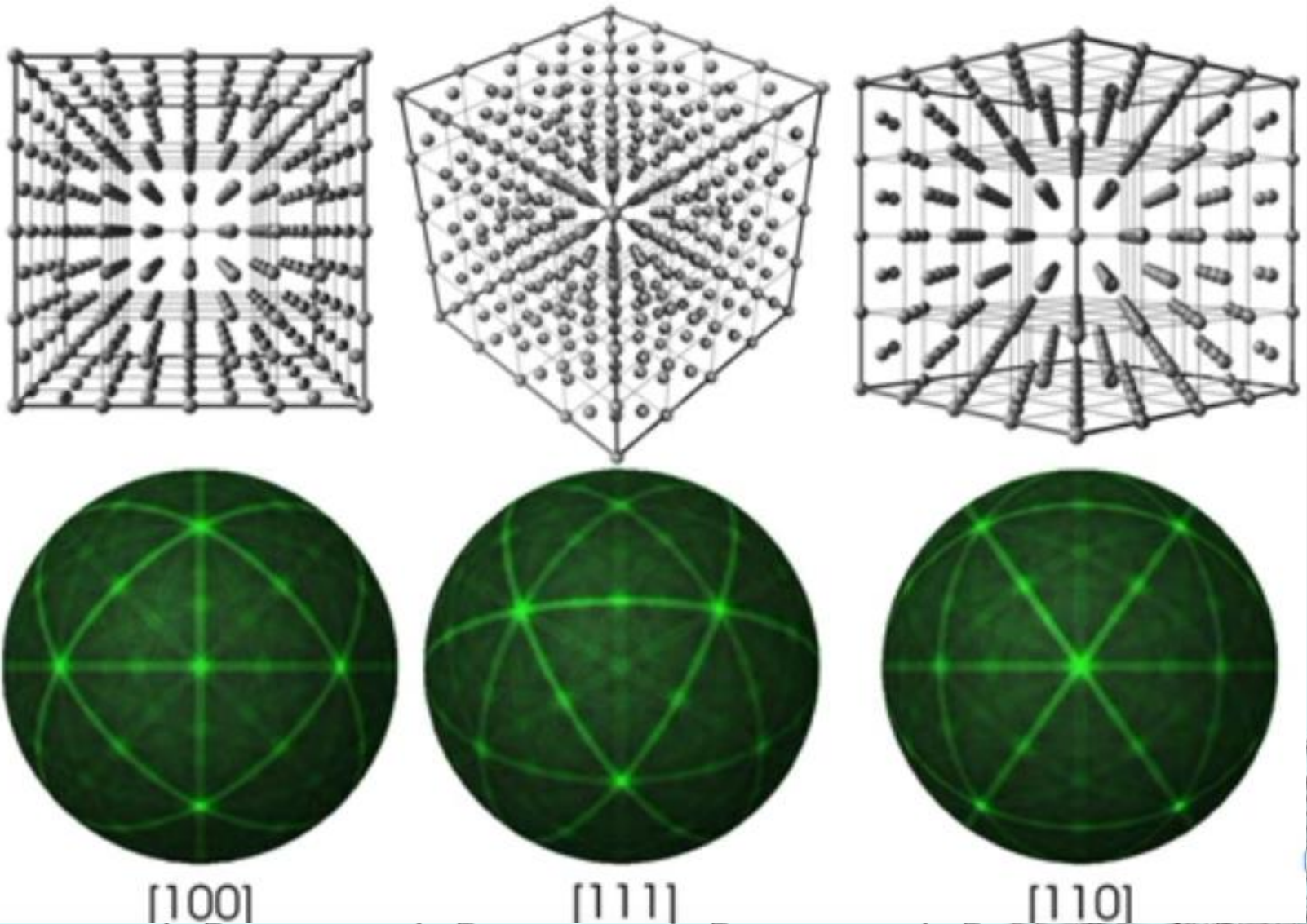
**Kikuchi patterns**



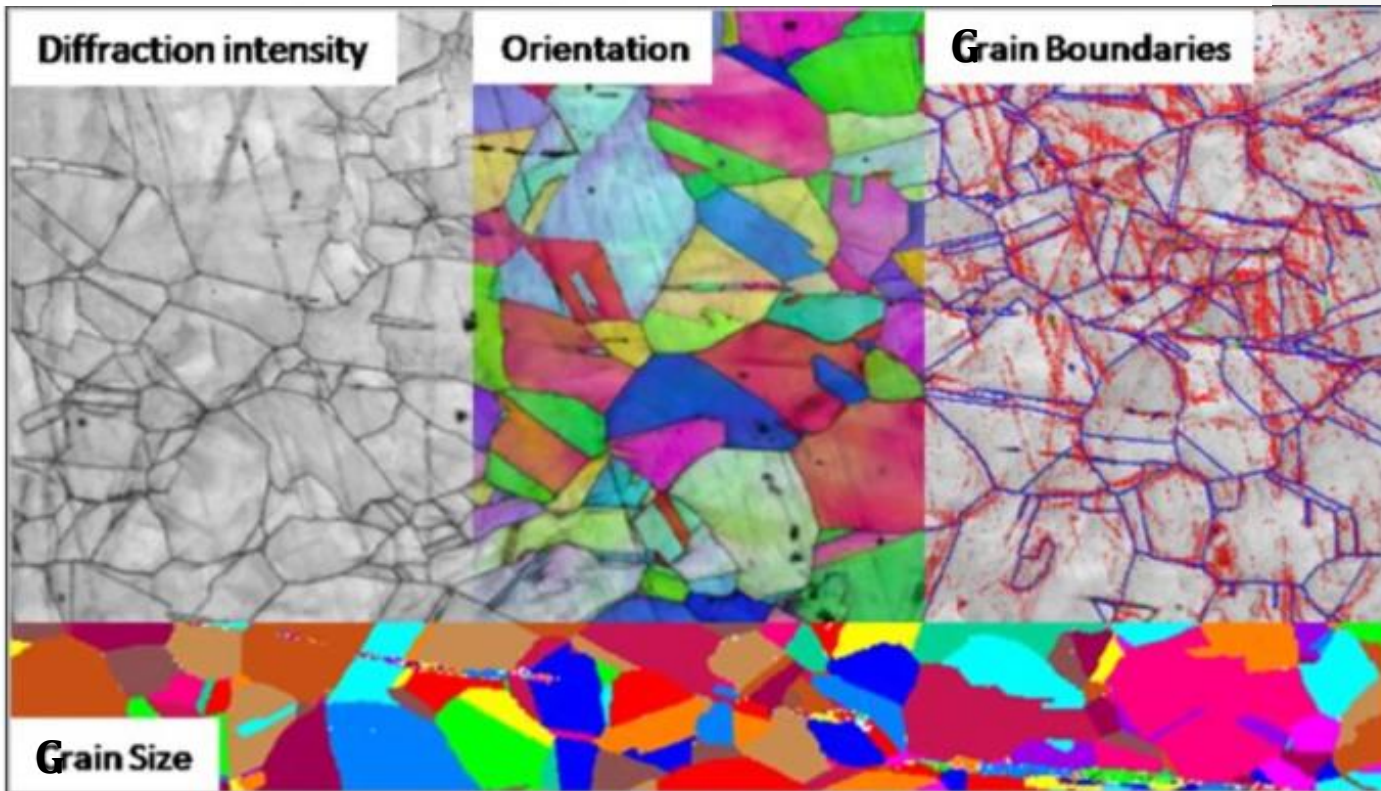
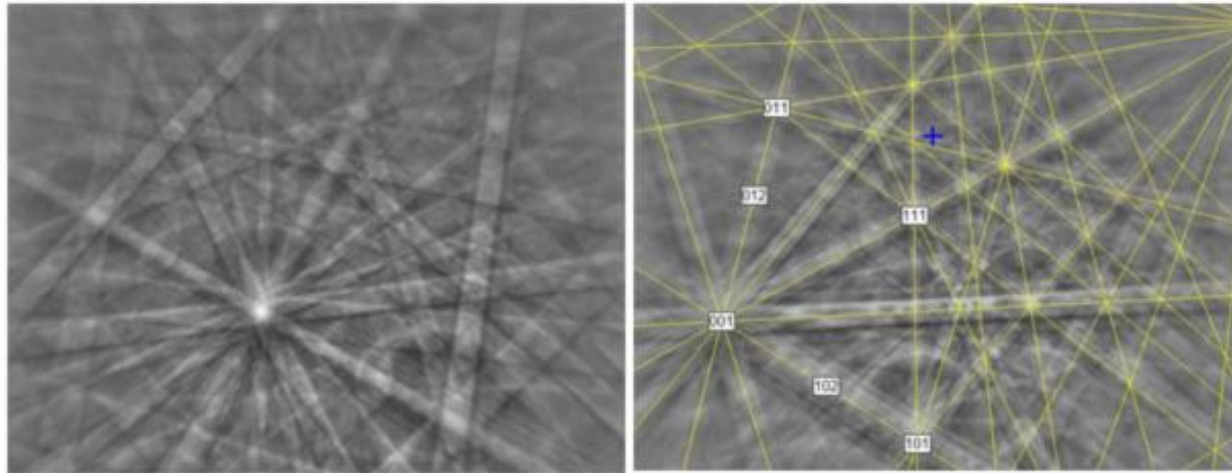
Diffracted electrons form cone which intersects phosphor screen



# Kikuchi patterns for different planes



# EBSD of AlN



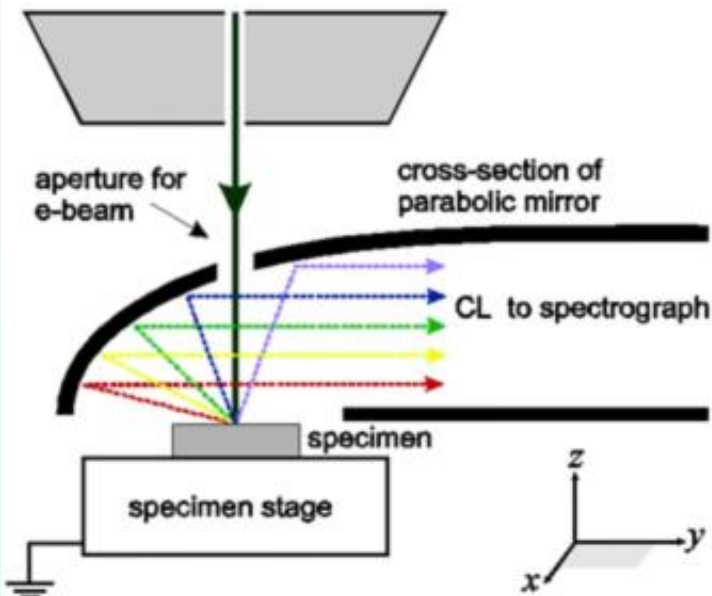


# Cathodoluminescence in the SEM

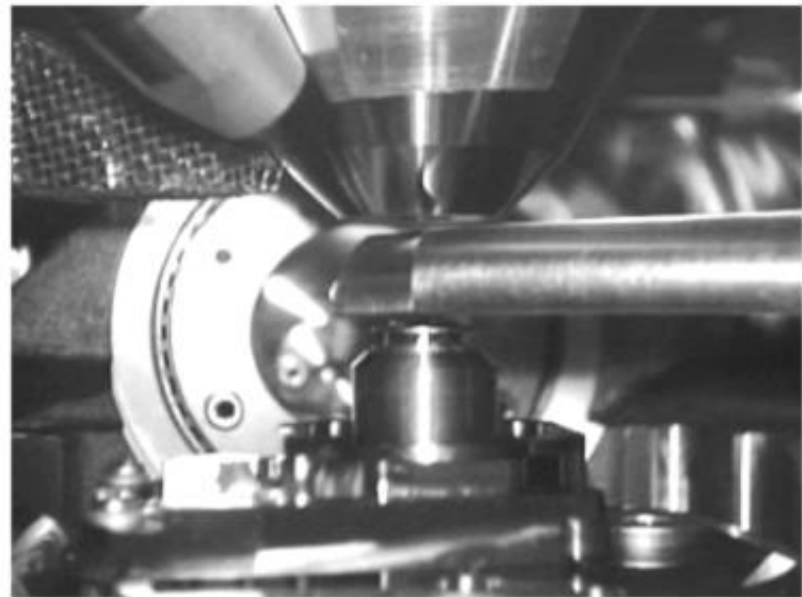
**NOT REQUIRED**

- Emission of light from a material during irradiation by an energetic beam of electrons
  - 300-1700nm optical range
  - monochromatic imaging with (0.1-1.0 $\mu$ m spatial resolution)
  - applications include semiconductor and defect analysis, quantum wells, plasmons
- Collection of emitted light using a paraboloidal mirror

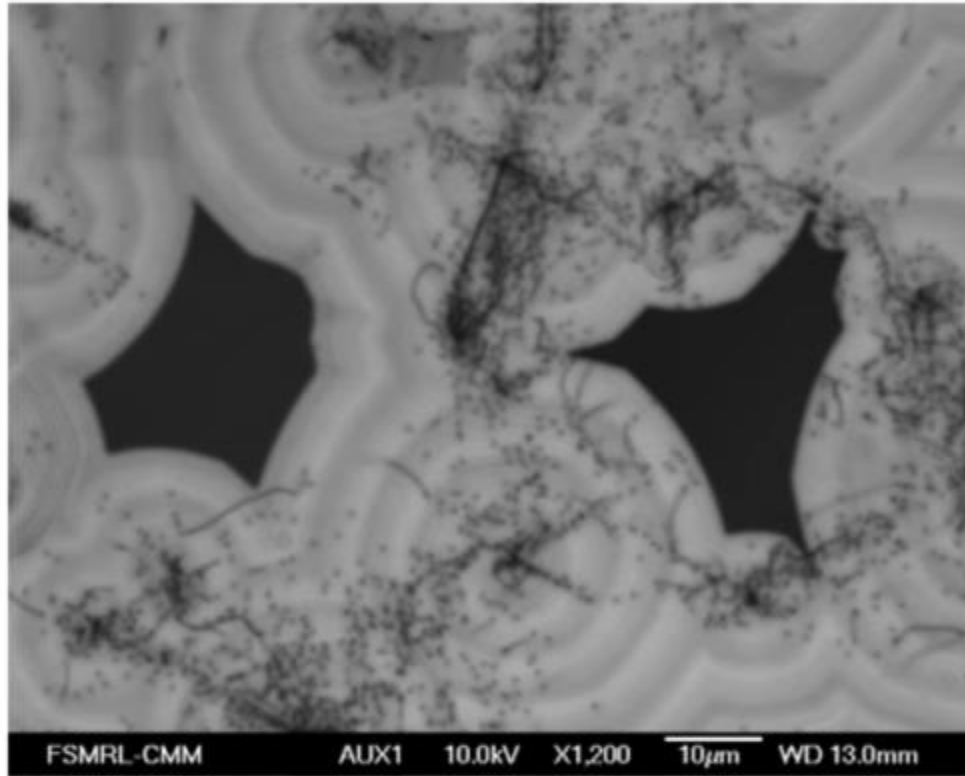
Schematic of CL system



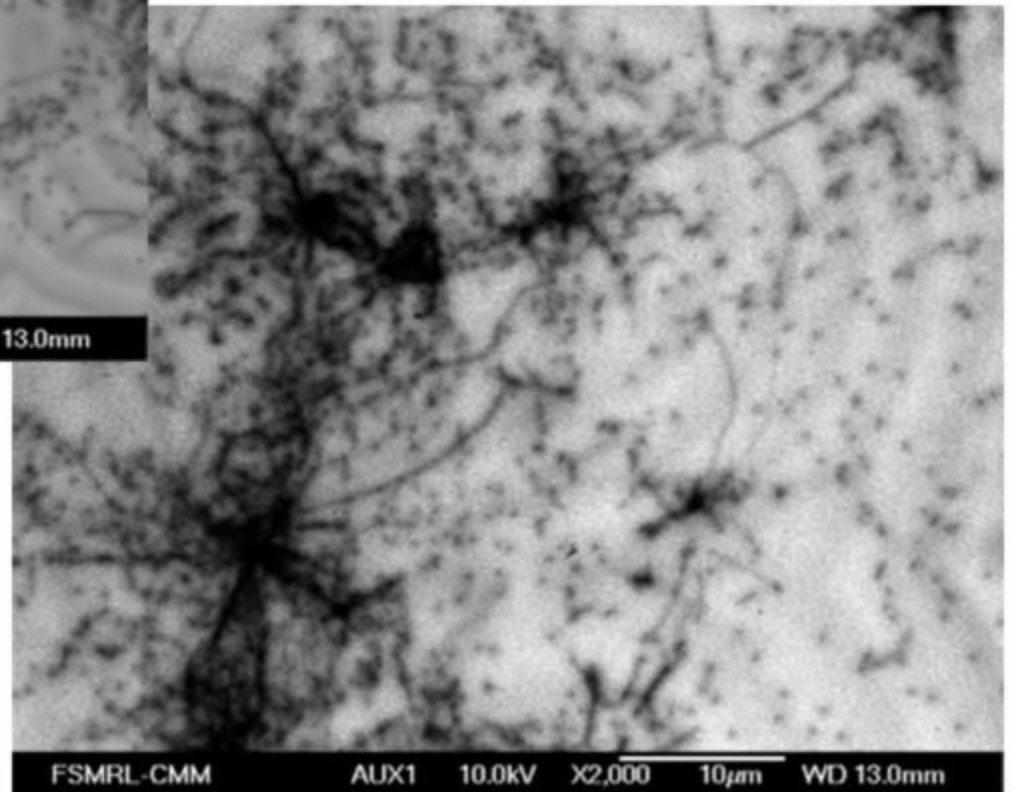
Collection optic in position between OL lens and sample



# Panchromatic CL Imaging GaN



Defects (dislocations) are observed as points or lines of reduced emission; act as e-h pair traps with non-radiative recombination





# Remarks for Backscattered Electrons

## Some of the properties of backscattered electrons

- The signal intensity increases with the increasing of the  $Z$  of the sample. Elements with higher  $Z$  appear brighter on the SEM screen.
- No significant changes in the backscattered emissions with the increase in acceleration voltage.
- Backscattered electron emissions from the specimen increase with the tilt angle.
- BSE are mostly affected by subsurface features of the specimen structure, and they are not used for topography analysis of the sample surface. They can be used for imaging purposes but without fine resolution at high magnifications.

# **Video demo**

**Video 1: SE and BSE imaging**

**Video 2: EBSD**

